

Supplementary Material for Linear Ordering Problem: Time for a Change

Fabrizio Fagiolo¹, Marco Baiocchi², Valentino Santucci¹

¹ University for Foreigners of Perugia

² University of Perugia

¹ f.fagiolo@phd.unistrapg.it, valentino.santucci@unistrapg.it

² marco.baiocchi@unipg.it

March 2026

In this document we provide the following data.

- Tables 1–3 summarize the results for the single-solution setting across the considered instance sets (**rxr**, **pxp**, **os300**). The symbols $\blacktriangle$ and ∇ indicate that the algorithm performs statistically better or worse, respectively, than its competitor, while the absence of a symbol denotes no significant difference.
- Tables 4–30 present the results for the multi-solution setting, with one table corresponding to each combination of instance set (**rxr**, **pxp**, **os300**), archive size (m), and metric (Φ , Δ_{NN} , Δ_{SP}). The symbols $\blacktriangle$ and ∇ indicate that the algorithm performs statistically better or worse, respectively, than its competitor, while the absence of a symbol denotes no significant difference.
- Figures 1–18 present the scatter plots, both “ Φ vs. Δ_{NN} ” and “ Φ vs. Δ_{SP} ”, for each single instance, where each point corresponds to the solution set obtained by a single execution of the algorithms.

Statistical significance in all tables is assessed using the Mann-Whitney U test with a significance level of 0.05, and p-values are adjusted using the Benjamini-Hochberg procedure.

Table 1: Results for classical setting (LOP) on os300 instances

Instance	CD-RVNS			MA-EDM		
	Best	Median	IQR	Best	Median	IQR
os300_at_2022_n300	28243679	28175026 ▽	35489	28243679	28228057 ▲	11138
os300_au_2022_n300	42091437	41979994 ▽	28556	42091437	42004540 ▲	40877
os300_be_2022_n300	26563205	26514166 ▽	28571	26563205	26527532 ▲	19636
os300_bg_2022_n300	36423853	36355238 ▽	39376	36423853	36385711 ▲	36896
os300_br_2022_n300	38581839	38457292 ▽	67950	38581839	38537140 ▲	35959
os300_ca_2022_n300	38445729	38373506 ▽	47844	38445729	38403718 ▲	53109
os300_ch_2022_n300	33018980	32896807 ▽	38510	33018980	32972636 ▲	32189
os300_cn_2022_n300	56798580	56632832 ▽	51386	56798580	56695628 ▲	53058
os300_cy_2022_n300	39300450	39168943 ▽	71416	39300450	39263457 ▲	27214
os300_cz_2022_n300	27123568	27061229 ▽	32171	27123568	27093757 ▲	14081
os300_de_2022_n300	26935500	26857898 ▽	26037	26935500	26915364 ▲	18340
os300_dk_2022_n300	35215033	35135597 ▽	53510	35215033	35156434 ▲	31932
os300_ee_2022_n300	33789498	33687218 ▽	37548	33789498	33750548 ▲	28746
os300_es_2022_n300	26643465	26567355 ▽	31320	26643465	26611429 ▲	20842
os300_fi_2022_n300	31560107	31487729 ▽	41325	31560107	31512874 ▲	22698
os300_fr_2022_n300	32127336	32046909 ▽	44052	32127336	32097975 ▲	41493
os300_gb_2022_n300	29553210	29487485 ▽	37957	29553210	29522823 ▲	23192
os300_gr_2022_n300	30984307	30906246 ▽	29079	30984307	30953620 ▲	10650
os300_hr_2022_n300	30687985	30602148 ▽	40126	30687985	30646139 ▲	24215
os300_hu_2022_n300	28960131	28885484 ▽	41621	28960131	28935349 ▲	24853
os300_id_2022_n300	64792455	64549234 ▽	141524	64792455	64724542 ▲	69094
os300_ie_2022_n300	23356772	23299913 ▽	37344	23356772	23327833 ▲	15694
os300_in_2022_n300	38414884	38271902 ▽	68465	38414884	38376672 ▲	33889
os300_it_2022_n300	27239986	27142874 ▽	39739	27239986	27190512 ▲	30394
os300_jp_2022_n300	37384329	37267367 ▽	74377	37384329	37337370 ▲	23884
os300_kr_2022_n300	34421850	34324784 ▽	40698	34421850	34387583 ▲	38809
os300_lt_2022_n300	28023381	27947262 ▽	18683	28023381	27986386 ▲	29352
os300_lu_2022_n300	31888525	31844253	33087	31888525	31841426	24743
os300_lv_2022_n300	32935749	32881366 ▽	29216	32935749	32902726 ▲	20101
os300_mt_2022_n300	45500160	45400004 ▽	85920	45500160	45438546 ▲	42388
os300_mx_2022_n300	33047210	32966544 ▽	54977	33047210	33005516 ▲	30371
os300_nl_2022_n300	21768835	21722606 ▽	27205	21768835	21760584 ▲	8528
os300_no_2022_n300	49039776	48860713 ▽	89084	49039776	48989826 ▲	59637
os300_pl_2022_n300	23799046	23751658 ▽	18437	23799046	23774682 ▲	15548
os300_pt_2022_n300	27235539	27161224 ▽	27593	27235539	27216574 ▲	21585
os300_ro_2022_n300	37256782	37114276 ▽	90723	37256782	37172353 ▲	44869
os300_ru_2022_n300	40609705	40528856 ▽	80311	40609705	40581006 ▲	16955
os300_se_2022_n300	25872116	25805400 ▽	27063	25872116	25842856 ▲	15568
os300_si_2022_n300	31694055	31626946 ▽	52095	31694055	31665518 ▲	15574
os300_sk_2022_n300	28402484	28314890 ▽	33192	28402484	28374928 ▲	26401
os300_tr_2022_n300	34418550	34339752 ▽	23760	34418550	34382844 ▲	27236
os300_tw_2022_n300	66015071	65878610 ▽	135368	66015071	65968872 ▲	50450
os300_us_2022_n300	37101549	36983252 ▽	68319	37101549	37058007 ▲	39701
os300_wa_2022_n300	15455699	15412038 ▽	16449	15455699	15433734 ▲	11498
os300_we_2022_n300	26505810	26469742	27542	26505810	26475162	21865
os300_wf_2022_n300	32611507	32557080	31336	32611507	32546784	26882
os300_wl_2022_n300	24513154	24447462 ▽	34520	24513154	24480604 ▲	13322
os300_wm_2022_n300	16166472	16116169 ▽	20874	16166472	16145740 ▲	13724
os300_za_2022_n300	48803579	48650110 ▽	91252	48803579	48764856 ▲	33564

Table 2: Results for classical setting (LOP) on pxp instances

Instance	CD-RVNS			MA-EDM		
	Best	Median	IQR	Best	Median	IQR
pxp_at_2022_n155	4997262	4997262 ▲	0	4997262	4997252 ▽	68
pxp_au_2022_n155	6509976	6509882	94	6509976	6509970	74
pxp_be_2022_n155	4840309	4840309	0	4840309	4840309	66
pxp_bg_2022_n135	5550530	5550359	128	5550530	5550478	186
pxp_br_2022_n155	5993534	5993056 ▽	230	5993534	5993152 ▲	317
pxp_ca_2022_n158	5607426	5607426	0	5607426	5607426	0
pxp_ch_2022_n141	4873413	4873413	0	4873413	4873413	0
pxp_cn_2022_n169	7846651	7846651 ▲	0	7846651	7846056 ▽	861
pxp_cy_2022_n116	3473529	3473529 ▲	0	3473529	3473496 ▽	131
pxp_cz_2022_n162	6076090	6076090	0	6076090	6076090	0
pxp_de_2022_n165	5833705	5833280 ▽	0	5833705	5833705 ▲	4
pxp_dk_2022_n141	4795237	4795237 ▲	0	4795237	4795235 ▽	2
pxp_ee_2022_n126	4271813	4271813	0	4271813	4271813	0
pxp_es_2022_n162	5505970	5505970	0	5505970	5505970	61
pxp_fi_2022_n153	5823310	5823310	1	5823310	5823310	0
pxp_fr_2022_n164	6191534	6191534 ▲	31	6191534	6191503 ▽	36
pxp_gb_2022_n159	5322836	5322836 ▲	0	5322836	5322692 ▽	50
pxp_gr_2022_n130	3400458	3400458	0	3400458	3400458	0
pxp_hr_2022_n140	4686061	4686061	0	4686061	4686061	0
pxp_hu_2022_n160	5739389	5739235 ▽	4	5739389	5739389 ▲	3
pxp_id_2022_n132	5249010	5249010 ▲	0	5249010	5248495 ▽	528
pxp_ie_2022_n136	2351111	2351111	0	2351111	2351111	0
pxp_in_2022_n145	5026649	5026649 ▲	0	5026649	5026629 ▽	5
pxp_it_2022_n165	5986644	5986055 ▲	303	5986644	5986020 ▽	174
pxp_jp_2022_n159	6439663	6439663 ▲	0	6439663	6439659 ▽	4
pxp_kr_2022_n149	5254433	5254433 ▲	0	5254433	5253929 ▽	470
pxp_lt_2022_n133	3695421	3695421	0	3695421	3695421	0
pxp_lu_2022_n120	2534779	2534779 ▲	0	2534779	2534748 ▽	78
pxp_lv_2022_n132	4580756	4580756	0	4580756	4580756	0
pxp_mt_2022_n89	2345841	2345841	0	2345841	2345841	0
pxp_mx_2022_n139	4176737	4176712 ▲	270	4176737	4176446 ▽	0
pxp_nl_2022_n158	4248155	4248155	0	4248155	4248155	76
pxp_no_2022_n145	6631865	6631865 ▲	0	6631865	6631827 ▽	13
pxp_pl_2022_n166	5465401	5465401	0	5465401	5465401	24
pxp_pt_2022_n149	4974639	4974639 ▲	0	4974639	4973856 ▽	585
pxp_ro_2022_n151	5449486	5449486 ▲	0	5449486	5448906 ▽	631
pxp_ru_2022_n136	5230190	5230190 ▲	0	5230190	5230056 ▽	724
pxp_se_2022_n156	5079306	5079262 ▽	44	5079306	5079306 ▲	0
pxp_si_2022_n129	4262916	4262916	0	4262916	4262916	0
pxp_sk_2022_n151	5016909	5016909	0	5016909	5016909	746
pxp_tr_2022_n145	5365578	5365578 ▲	0	5365578	5365433 ▽	177
pxp_tw_2022_n138	5532757	5532757 ▲	0	5532757	5532556 ▽	156
pxp_us_2022_n170	6613891	6613891	0	6613891	6613891	0
pxp_wa_2022_n176	4011427	4011427	0	4011427	4011427	0
pxp_we_2022_n144	4492970	4492970 ▲	0	4492970	4492856 ▽	28
pxp_wf_2022_n141	4829905	4829905	0	4829905	4829905	0
pxp_wl_2022_n164	4966516	4966516 ▲	0	4966516	4966480 ▽	28
pxp_wm_2022_n161	3485302	3485302 ▲	0	3485302	3485292 ▽	47
pxp_za_2022_n134	5634755	5634755 ▲	0	5634755	5634406 ▽	799

Table 3: Results for classical setting (LOP) on rxr instances

Instance	CD-RVNS			MA-EDM		
	Best	Median	IQR	Best	Median	IQR
rxr_1995_n49	305741	305741	0	305741	305741	0
rxr_1996_n49	321479	321479	0	321479	321479	0
rxr_1997_n49	344182	344182	0	344182	344182	0
rxr_1998_n49	341390	341390	0	341390	341390	0
rxr_1999_n49	339587	339587	0	339587	339587	0
rxr_2000_n49	366789	366789	0	366789	366789	0
rxr_2001_n49	366155	366155	0	366155	366155	0
rxr_2002_n49	359952	359952	0	359952	359952	0
rxr_2003_n49	342326	342326	0	342326	342326	0
rxr_2004_n49	361744	361744	0	361744	361744	0
rxr_2005_n49	374491	374491	0	374491	374491	0
rxr_2006_n49	389927	389927	0	389927	389927	0
rxr_2007_n49	384053	384053	0	384053	384053	0
rxr_2008_n49	393668	393668	0	393668	393668	0
rxr_2009_n49	353109	353109	0	353109	353109	0
rxr_2010_n49	409621	409621	0	409621	409621	0
rxr_2011_n49	427225	427225	0	427225	427225	0
rxr_2012_n49	444306	444306	0	444306	444306	0
rxr_2013_n49	441753	441753	0	441753	441753	0
rxr_2014_n49	439250	439250	0	439250	439250	0
rxr_2015_n49	443549	443549	0	443549	443549	0
rxr_2016_n49	445805	445805	0	445805	445805	0
rxr_2017_n49	444129	444129	0	444129	444129	0
rxr_2018_n49	462936	462936	0	462936	462936	0
rxr_2019_n49	465593	465593	0	465593	465593	0
rxr_2020_n49	430104	430104	0	430104	430104	0
rxr_2021_n49	431939	431939	0	431939	431939	0
rxr_2022_n49	450783	450783	0	450783	450783	0

Table 4: Results on MS-LOP for Φ metric and $m = 5$ on os300 instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
os300_at_2022_n300	5	28243679	28175009 ▽	35495	28243679	28228047 ▲	11175
os300_au_2022_n300	5	42091432	41979994 ▽	28543	41983708	42004520 ▲	40881
os300_be_2022_n300	5	26563201	26514132 ▽	28558	26510456	26527437 ▲	19639
os300_bg_2022_n300	5	36423843	36355169 ▽	39284	36339035	36385643 ▲	36871
os300_br_2022_n300	5	38581789	38457272 ▽	67681	38526322	38537049 ▲	35938
os300_ca_2022_n300	5	38445653	38373468 ▽	47869	38359566	38403662 ▲	53095
os300_ch_2022_n300	5	33018970	32896789 ▽	38505	32983236	32972611 ▲	32160
os300_cn_2022_n300	5	56798511	56632832 ▽	51375	56694397	56695604 ▲	53063
os300_cy_2022_n300	5	39300444	39168914 ▽	71371	39267694	39263420 ▲	27243
os300_cz_2022_n300	5	27123566	27061214 ▽	32316	27112412	27093747 ▲	14053
os300_de_2022_n300	5	26935421	26857884 ▽	26035	26914138	26915310 ▲	18137
os300_dk_2022_n300	5	35214798	35135586 ▽	53680	35214798	35156414 ▲	31950
os300_ee_2022_n300	5	33789490	33687205 ▽	37512	33733094	33750484 ▲	28732
os300_es_2022_n300	5	26643438	26567316 ▽	31334	26605372	26611385 ▲	20940
os300_fi_2022_n300	5	31560086	31487716 ▽	41355	31556490	31512850 ▲	22717
os300_fr_2022_n300	5	32127313	32046868 ▽	44061	32084725	32097949 ▲	41377
os300_gb_2022_n300	5	29553191	29487442 ▽	38009	29531833	29522792 ▲	23172
os300_gr_2022_n300	5	30984288	30906183 ▽	29104	30933491	30953564 ▲	10688
os300_hr_2022_n300	5	30687973	30602106 ▽	40144	30639284	30646131 ▲	24216
os300_hu_2022_n300	5	28960052	28885483 ▽	41613	28953326	28935330 ▲	24764
os300_id_2022_n300	5	64792430	64549231 ▽	141530	64554655	64724519 ▲	69090
os300_ie_2022_n300	5	23356772	23299912 ▽	37338	23299556	23327833 ▲	15633
os300_in_2022_n300	5	38414857	38271870 ▽	68456	38407153	38376665 ▲	33908
os300_it_2022_n300	5	27239961	27142859 ▽	39746	27168478	27190386 ▲	30410
os300_jp_2022_n300	5	37384325	37267271 ▽	74417	37296035	37337360 ▲	23674
os300_kr_2022_n300	5	34421848	34324781 ▽	40677	34372482	34387568 ▲	38778
os300_lt_2022_n300	5	28023369	27947241 ▽	18691	28013722	27986381 ▲	29359
os300_lu_2022_n300	5	31888475	31844253	33087	31861759	31841391	24740
os300_lv_2022_n300	5	32935645	32881354 ▽	29197	32885892	32902640 ▲	20084
os300_mt_2022_n300	5	45499608	45399991 ▽	85833	45459203	45438495 ▲	42464
os300_mx_2022_n300	5	33047210	32966532 ▽	54971	32988830	33005509 ▲	30254
os300_nl_2022_n300	5	21768812	21722590 ▽	27216	21764404	21760552 ▲	8540
os300_no_2022_n300	5	49039774	48860684 ▽	89081	48972884	48989808 ▲	59695
os300_pl_2022_n300	5	23799044	23751637 ▽	18469	23768624	23774650 ▲	15544
os300_pt_2022_n300	5	27235473	27161209 ▽	27551	27218277	27216548 ▲	21632
os300_ro_2022_n300	5	37256768	37114250 ▽	90662	37243798	37172277 ▲	44920
os300_ru_2022_n300	5	40609696	40528844 ▽	80282	40597201	40580958 ▲	16947
os300_se_2022_n300	5	25872051	25805342 ▽	27091	25847481	25842839 ▲	15522
os300_si_2022_n300	5	31694025	31626939 ▽	52106	31675855	31665467 ▲	15557
os300_sk_2022_n300	5	28402424	28314846 ▽	33300	28388013	28374912 ▲	26195
os300_tr_2022_n300	5	34418462	34339742 ▽	23768	34372054	34382724 ▲	27280
os300_tw_2022_n300	5	66015040	65878608 ▽	135410	66012988	65968731 ▲	50433
os300_us_2022_n300	5	37101531	36983233 ▽	68358	37079504	37057946 ▲	39711
os300_wa_2022_n300	5	15455675	15412027 ▽	16444	15438206	15433716 ▲	11520
os300_we_2022_n300	5	26505803	26469735	27523	26479942	26475099	21857
os300_wf_2022_n300	5	32611502	32557077	31331	32512261	32546755	26873
os300_wl_2022_n300	5	24513152	24447453 ▽	34528	24465187	24480600 ▲	13322
os300_wm_2022_n300	5	16166462	16116063 ▽	20879	16145668	16145730 ▲	13706
os300_za_2022_n300	5	48803453	48650051 ▽	91246	48793050	48764776 ▲	33780

Table 5: Results on MS-LOP for Φ metric and $m = 10$ on os300 instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
os300_at_2022_n300	10	28243679	28174989 ▽	35525	28243674	28228032 ▲	11177
os300_au_2022_n300	10	42091432	41979968 ▽	28532	41983638	42004503 ▲	40877
os300_be_2022_n300	10	26563201	26514103 ▽	28544	26510450	26527346 ▲	19672
os300_bg_2022_n300	10	36423843	36355112 ▽	39226	36338895	36385543 ▲	36877
os300_br_2022_n300	10	38581789	38457241 ▽	67602	38526271	38536952 ▲	35918
os300_ca_2022_n300	10	38445653	38373434 ▽	47902	38359558	38403625 ▲	53089
os300_ch_2022_n300	10	33018970	32896752 ▽	38471	32983198	32972577 ▲	32156
os300_cn_2022_n300	10	56798511	56632830 ▽	51332	56694356	56695563 ▲	53070
os300_cy_2022_n300	10	39300444	39168898 ▽	71333	39267689	39263384 ▲	27270
os300_cz_2022_n300	10	27123566	27061191 ▽	32385	27112337	27093731 ▲	14037
os300_de_2022_n300	10	26935421	26857869 ▽	26036	26913998	26915195 ▲	17853
os300_dk_2022_n300	10	35214798	35135568 ▽	53834	35214707	35156389 ▲	31986
os300_ee_2022_n300	10	33789490	33687189 ▽	37459	33732973	33750341 ▲	28766
os300_es_2022_n300	10	26643438	26567264 ▽	31350	26605354	26611338 ▲	20972
os300_fi_2022_n300	10	31560086	31487692 ▽	41380	31556462	31512796 ▲	22736
os300_fr_2022_n300	10	32127313	32046808 ▽	44037	32084676	32097909 ▲	41367
os300_gb_2022_n300	10	29553191	29487394 ▽	38044	29531802	29522725 ▲	23164
os300_gr_2022_n300	10	30984288	30906116 ▽	29169	30933402	30953508 ▲	10761
os300_hr_2022_n300	10	30687973	30602076 ▽	40149	30639264	30646105 ▲	24224
os300_hu_2022_n300	10	28960052	28885468 ▽	41564	28953271	28935306 ▲	24698
os300_id_2022_n300	10	64792430	64549220 ▽	141599	64554647	64724485 ▲	69051
os300_ie_2022_n300	10	23356772	23299906 ▽	37336	23299554	23327829 ▲	15582
os300_in_2022_n300	10	38414857	38271840 ▽	68422	38407139	38376652 ▲	33916
os300_it_2022_n300	10	27239961	27142834 ▽	39775	27168239	27190293 ▲	30422
os300_jp_2022_n300	10	37384325	37267212 ▽	74427	37295866	37337350 ▲	23574
os300_kr_2022_n300	10	34421848	34324773 ▽	40623	34372321	34387548 ▲	38750
os300_lt_2022_n300	10	28023369	27947205 ▽	18690	28013684	27986305 ▲	29361
os300_lu_2022_n300	10	31888475	31844244	33082	31861727	31841349	24735
os300_lv_2022_n300	10	32935645	32881321 ▽	29192	32885843	32902536 ▲	20091
os300_mt_2022_n300	10	45499608	45399964 ▽	85730	45459192	45438456 ▲	42441
os300_mx_2022_n300	10	33047210	32966516 ▽	54939	32988717	33005496 ▲	30180
os300_nl_2022_n300	10	21768812	21722538 ▽	27222	21764363	21760517 ▲	8589
os300_no_2022_n300	10	49039774	48860623 ▽	89174	48972858	48989780 ▲	59780
os300_pl_2022_n300	10	23799044	23751618 ▽	18497	23768501	23774627 ▲	15534
os300_pt_2022_n300	10	27235473	27161169 ▽	27516	27218254	27216513 ▲	21710
os300_ro_2022_n300	10	37256768	37114234 ▽	90653	37243761	37172149 ▲	44916
os300_ru_2022_n300	10	40609696	40528829 ▽	80268	40597182	40580910 ▲	16934
os300_se_2022_n300	10	25872051	25805264 ▽	27108	25847450	25842816 ▲	15480
os300_si_2022_n300	10	31694025	31626915 ▽	52125	31675813	31665436 ▲	15561
os300_sk_2022_n300	10	28402424	28314811 ▽	33461	28387936	28374890 ▲	26070
os300_tr_2022_n300	10	34418462	34339727 ▽	23786	34372001	34382660 ▲	27407
os300_tw_2022_n300	10	66015040	65878567 ▽	135338	66012889	65968611 ▲	50410
os300_us_2022_n300	10	37101531	36983210 ▽	68392	37079487	37057910 ▲	39722
os300_wa_2022_n300	10	15455675	15412012 ▽	16443	15438202	15433691 ▲	11538
os300_we_2022_n300	10	26505803	26469713	27499	26479936	26475060	21854
os300_wf_2022_n300	10	32611502	32557070	31130	32512238	32546732	26862
os300_wl_2022_n300	10	24513152	24447443 ▽	34521	24465160	24480591 ▲	13322
os300_wm_2022_n300	10	16166462	16116017 ▽	20889	16145629	16145704 ▲	13692
os300_za_2022_n300	10	48803453	48650003 ▽	91248	48793010	48764724 ▲	33876

Table 6: Results on MS-LOP for Φ metric and $m = 15$ on os300 instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
os300_at_2022_n300	15	28243679	28174972 ▽	35538	28243659	28228019 ▲	11173
os300_au_2022_n300	15	42091432	41979940 ▽	28507	41983596	42004490 ▲	40874
os300_be_2022_n300	15	26563201	26514077 ▽	28534	26510443	26527274 ▲	19705
os300_bg_2022_n300	15	36423843	36355056 ▽	39182	36338785	36385253 ▲	36892
os300_br_2022_n300	15	38581789	38457204 ▽	67566	38526238	38536887 ▲	35920
os300_ca_2022_n300	15	38445653	38373406 ▽	47924	38359550	38403592 ▲	53105
os300_ch_2022_n300	15	33018970	32896717 ▽	38455	32983179	32972554 ▲	32156
os300_cn_2022_n300	15	56798511	56632829 ▽	51291	56694328	56695527 ▲	53086
os300_cy_2022_n300	15	39300444	39168886 ▽	71321	39267681	39263325 ▲	27362
os300_cz_2022_n300	15	27123566	27061167 ▽	32422	27112280	27093715 ▲	14032
os300_de_2022_n300	15	26935421	26857857 ▽	26026	26913912	26915120 ▲	17684
os300_dk_2022_n300	15	35214798	35135554 ▽	53997	35214620	35156368 ▲	32002
os300_ee_2022_n300	15	33789490	33687171 ▽	37464	33732889	33750240 ▲	28800
os300_es_2022_n300	15	26643438	26567219 ▽	31356	26605334	26611311 ▲	21010
os300_fi_2022_n300	15	31560086	31487672 ▽	41395	31556438	31512758 ▲	22754
os300_fr_2022_n300	15	32127313	32046758 ▽	44006	32084611	32097870 ▲	41368
os300_gb_2022_n300	15	29553191	29487360 ▽	38061	29531771	29522664 ▲	23152
os300_gr_2022_n300	15	30984288	30906053 ▽	29225	30933339	30953466 ▲	10798
os300_hr_2022_n300	15	30687973	30602043 ▽	40185	30639242	30646076 ▲	24217
os300_hu_2022_n300	15	28960052	28885450 ▽	41536	28953226	28935284 ▲	24649
os300_id_2022_n300	15	64792430	64549188 ▽	141631	64554636	64724458 ▲	69016
os300_ie_2022_n300	15	23356772	23299886 ▽	37337	23299545	23327825 ▲	15550
os300_in_2022_n300	15	38414857	38271806 ▽	68396	38407132	38376639 ▲	33916
os300_it_2022_n300	15	27239961	27142812 ▽	39782	27168027	27190239 ▲	30434
os300_jp_2022_n300	15	37384325	37267144 ▽	74438	37295776	37337319 ▲	23530
os300_kr_2022_n300	15	34421848	34324764 ▽	40596	34372042	34387527 ▲	38735
os300_lt_2022_n300	15	28023369	27947180 ▽	18696	28013646	27986229 ▲	29362
os300_lu_2022_n300	15	31888475	31844231	33081	31861684	31841316	24727
os300_lv_2022_n300	15	32935645	32881278 ▽	29182	32885807	32902428 ▲	20090
os300_mt_2022_n300	15	45499608	45399925 ▽	85637	45459180	45438424 ▲	42413
os300_mx_2022_n300	15	33047210	32966504 ▽	54922	32988630	33005480 ▲	30149
os300_nl_2022_n300	15	21768812	21722492 ▽	27220	21764337	21760483 ▲	8637
os300_no_2022_n300	15	49039774	48860560 ▽	89226	48972843	48989759 ▲	59863
os300_pl_2022_n300	15	23799044	23751604 ▽	18515	23768426	23774607 ▲	15536
os300_pt_2022_n300	15	27235473	27161135 ▽	27504	27218232	27216486 ▲	21738
os300_ro_2022_n300	15	37256768	37114222 ▽	90661	37243693	37172044 ▲	44888
os300_ru_2022_n300	15	40609696	40528815 ▽	80273	40597165	40580872 ▲	16912
os300_se_2022_n300	15	25872051	25805193 ▽	27113	25847424	25842790 ▲	15431
os300_si_2022_n300	15	31694025	31626893 ▽	52142	31675787	31665410 ▲	15551
os300_sk_2022_n300	15	28402424	28314784 ▽	33553	28387873	28374865 ▲	25971
os300_tr_2022_n300	15	34418462	34339715 ▽	23804	34371965	34382629 ▲	27448
os300_tw_2022_n300	15	66015040	65878536 ▽	135297	66012771	65968502 ▲	50377
os300_us_2022_n300	15	37101531	36983188 ▽	68419	37079473	37057846 ▲	39728
os300_wa_2022_n300	15	15455675	15411999 ▽	16441	15438197	15433674 ▲	11553
os300_we_2022_n300	15	26505803	26469675	27483	26479930	26475028	21850
os300_wf_2022_n300	15	32611502	32557061	31039	32512228	32546715	26834
os300_wl_2022_n300	15	24513152	24447434 ▽	34509	24465136	24480576 ▲	13325
os300_wm_2022_n300	15	16166462	16115995 ▽	20896	16145600	16145683 ▲	13690
os300_za_2022_n300	15	48803453	48649948 ▽	91222	48792989	48764678 ▲	33912

Table 7: Results on MS-LOP for Φ metric and $m = 5$ on PXP instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
pxp_at_2022_n155	5	4997262	4997262	▲	0	4997262	4997252 ▽ 68
pxp_au_2022_n155	5	6509976	6509882		94	6509967	6509970 74
pxp_be_2022_n155	5	4840309	4840309		0	4840309	4840309 66
pxp_bg_2022_n135	5	5550530	5550359		128	5550478	5550478 186
pxp_br_2022_n155	5	5993534	5993056	▽	230	5993099	5993152 ▲ 317
pxp_ca_2022_n158	5	5607426	5607426		0	5607426	5607426 0
pxp_ch_2022_n141	5	4873413	4873413		0	4873413	4873413 0
pxp_cn_2022_n169	5	7846651	7846651	▲	0	7846389	7846056 ▽ 861
pxp_cy_2022_n116	5	3473529	3473529	▲	0	3473386	3473496 ▽ 131
pxp_cz_2022_n162	5	6076090	6076090		0	6076090	6076090 0
pxp_de_2022_n165	5	5833705	5833280	▽	0	5833705	5833705 ▲ 4
pxp_dk_2022_n141	5	4795237	4795237	▲	0	4795235	4795235 ▽ 2
pxp_ee_2022_n126	5	4271813	4271813		0	4271813	4271813 0
pxp_es_2022_n162	5	5505970	5505970		0	5505909	5505970 61
pxp_fi_2022_n153	5	5823310	5823310		1	5823309	5823310 0
pxp_fr_2022_n164	5	6191534	6191534	▲	31	6191529	6191503 ▽ 36
pxp_gb_2022_n159	5	5322836	5322836	▲	0	5322742	5322692 ▽ 50
pxp_gr_2022_n130	5	3400458	3400458		0	3400458	3400458 0
pxp_hr_2022_n140	5	4686061	4686061		0	4686061	4686061 0
pxp_hu_2022_n160	5	5739389	5739235	▽	4	5739389	5739389 ▲ 3
pxp_id_2022_n132	5	5249010	5249010	▲	0	5248855	5248495 ▽ 528
pxp_ie_2022_n136	5	2351111	2351111		0	2351111	2351111 0
pxp_in_2022_n145	5	5026649	5026649	▲	0	5026629	5026629 ▽ 5
pxp_it_2022_n165	5	5986644	5986055	▲	303	5986037	5986020 ▽ 174
pxp_jp_2022_n159	5	6439663	6439663	▲	0	6439659	6439659 ▽ 4
pxp_kr_2022_n149	5	5254433	5254433	▲	0	5254433	5253929 ▽ 470
pxp_lt_2022_n133	5	3695421	3695421		0	3695421	3695421 0
pxp_lu_2022_n120	5	2534779	2534779	▲	0	2534779	2534748 ▽ 78
pxp_lv_2022_n132	5	4580756	4580756		0	4580756	4580756 0
pxp_mt_2022_n89	5	2345841	2345841		0	2345841	2345841 0
pxp_mx_2022_n139	5	4176737	4176712	▲	270	4176446	4176446 ▽ 0
pxp_nl_2022_n158	5	4248155	4248155		0	4248155	4248155 76
pxp_no_2022_n145	5	6631865	6631865	▲	0	6631865	6631827 ▽ 13
pxp_pl_2022_n166	5	5465401	5465401		0	5465401	5465401 24
pxp_pt_2022_n149	5	4974639	4974639	▲	0	4974624	4973856 ▽ 585
pxp_ro_2022_n151	5	5449486	5449486	▲	0	5448898	5448906 ▽ 631
pxp_ru_2022_n136	5	5230190	5230190	▲	0	5229836	5230056 ▽ 724
pxp_se_2022_n156	5	5079306	5079262	▽	44	5079306	5079306 ▲ 0
pxp_si_2022_n129	5	4262916	4262916		0	4262916	4262916 0
pxp_sk_2022_n151	5	5016909	5016909		0	5016909	5016909 746
pxp_tr_2022_n145	5	5365578	5365578	▲	0	5365349	5365433 ▽ 177
pxp_tw_2022_n138	5	5532757	5532757	▲	0	5532672	5532556 ▽ 156
pxp_us_2022_n170	5	6613891	6613891		0	6613891	6613891 0
pxp_wa_2022_n176	5	4011427	4011427		0	4011427	4011427 0
pxp_we_2022_n144	5	4492970	4492970	▲	0	4492856	4492856 ▽ 28
pxp_wf_2022_n141	5	4829905	4829905		0	4829905	4829905 0
pxp_wl_2022_n164	5	4966516	4966516	▲	0	4966480	4966480 ▽ 28
pxp_wm_2022_n161	5	3485302	3485302	▲	0	3485251	3485292 ▽ 47
pxp_zs_2022_n134	5	5634755	5634755	▲	0	5634755	5634406 ▽ 799

Table 8: Results on MS-LOP for Φ metric and $m = 10$ on PXP instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
pxp_at_2022_n155	10	4997262	4997262 ▲	0	4997262	4997252 ▽	68
pxp_au_2022_n155	10	6509976	6509882	94	6509967	6509970	74
pxp_be_2022_n155	10	4840309	4840309	0	4840309	4840309	66
pxp_bg_2022_n135	10	5550530	5550359	128	5550478	5550478	186
pxp_br_2022_n155	10	5993534	5993056 ▽	230	5993099	5993152 ▲	317
pxp_ca_2022_n158	10	5607426	5607426	0	5607426	5607426	0
pxp_ch_2022_n141	10	4873413	4873413	0	4873413	4873413	0
pxp_cn_2022_n169	10	7846651	7846651 ▲	0	7846389	7846056 ▽	861
pxp_cy_2022_n116	10	3473529	3473529 ▲	0	3473386	3473496 ▽	131
pxp_cz_2022_n162	10	6076090	6076090	0	6076090	6076090	0
pxp_de_2022_n165	10	5833705	5833280 ▽	0	5833705	5833705 ▲	4
pxp_dk_2022_n141	10	4795237	4795237 ▲	0	4795235	4795235 ▽	2
pxp_ee_2022_n126	10	4271813	4271813	0	4271813	4271813	0
pxp_es_2022_n162	10	5505970	5505970	0	5505909	5505970	61
pxp_fi_2022_n153	10	5823310	5823310	1	5823309	5823310	0
pxp_fr_2022_n164	10	6191534	6191534 ▲	31	6191529	6191503 ▽	36
pxp_gb_2022_n159	10	5322836	5322836 ▲	0	5322742	5322692 ▽	50
pxp_gr_2022_n130	10	3400458	3400458	0	3400458	3400458	0
pxp_hr_2022_n140	10	4686061	4686061	0	4686061	4686061	0
pxp_hu_2022_n160	10	5739389	5739235 ▽	4	5739389	5739389 ▲	3
pxp_id_2022_n132	10	5249010	5249010 ▲	0	5248855	5248495 ▽	528
pxp_ie_2022_n136	10	2351111	2351111	0	2351111	2351111	0
pxp_in_2022_n145	10	5026649	5026649 ▲	0	5026629	5026629 ▽	5
pxp_it_2022_n165	10	5986644	5986055 ▲	303	5986037	5986020 ▽	174
pxp_jp_2022_n159	10	6439663	6439663 ▲	0	6439659	6439659 ▽	4
pxp_kr_2022_n149	10	5254433	5254433 ▲	0	5254433	5253929 ▽	470
pxp_lt_2022_n133	10	3695421	3695421	0	3695421	3695421	0
pxp_lu_2022_n120	10	2534779	2534779 ▲	0	2534779	2534748 ▽	78
pxp_lv_2022_n132	10	4580756	4580756	0	4580756	4580756	0
pxp_mt_2022_n89	10	2345841	2345841	0	2345841	2345841	0
pxp_mx_2022_n139	10	4176737	4176712 ▲	270	4176446	4176446 ▽	0
pxp_nl_2022_n158	10	4248155	4248155	0	4248155	4248155	76
pxp_no_2022_n145	10	6631865	6631865 ▲	0	6631865	6631827 ▽	13
pxp_pl_2022_n166	10	5465401	5465401	0	5465401	5465401	24
pxp_pt_2022_n149	10	4974639	4974639 ▲	0	4974624	4973856 ▽	585
pxp_ro_2022_n151	10	5449486	5449486 ▲	0	5448898	5448906 ▽	631
pxp_ru_2022_n136	10	5230190	5230190 ▲	0	5229836	5230056 ▽	724
pxp_se_2022_n156	10	5079306	5079262 ▽	44	5079306	5079306 ▲	0
pxp_si_2022_n129	10	4262916	4262916	0	4262916	4262916	0
pxp_sk_2022_n151	10	5016909	5016909	0	5016909	5016909	746
pxp_tr_2022_n145	10	5365578	5365578 ▲	0	5365349	5365433 ▽	177
pxp_tw_2022_n138	10	5532757	5532757 ▲	0	5532672	5532556 ▽	156
pxp_us_2022_n170	10	6613891	6613891	0	6613891	6613891	0
pxp_wa_2022_n176	10	4011427	4011427	0	4011427	4011427	0
pxp_we_2022_n144	10	4492970	4492970 ▲	0	4492856	4492856 ▽	28
pxp_wf_2022_n141	10	4829905	4829905	0	4829905	4829905	0
pxp_wl_2022_n164	10	4966516	4966516 ▲	0	4966480	4966480 ▽	28
pxp_wm_2022_n161	10	3485302	3485302 ▲	0	3485251	3485292 ▽	47
pxp_za_2022_n134	10	5634755	5634755 ▲	0	5634755	5634406 ▽	799

Table 9: Results on MS-LOP for Φ metric and $m = 15$ on PXP instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
pxp_at_2022_n155	15	4997262	4997262 ▲	0	4997262	4997252 ▽	68
pxp_au_2022_n155	15	6509976	6509882	94	6509967	6509970	74
pxp_be_2022_n155	15	4840309	4840309	0	4840309	4840309	66
pxp_bg_2022_n135	15	5550530	5550359	128	5550478	5550478	186
pxp_br_2022_n155	15	5993534	5993056 ▽	230	5993099	5993152 ▲	317
pxp_ca_2022_n158	15	5607426	5607426	0	5607426	5607426	0
pxp_ch_2022_n141	15	4873413	4873413	0	4873413	4873413	0
pxp_cn_2022_n169	15	7846651	7846651 ▲	0	7846389	7846056 ▽	861
pxp_cy_2022_n116	15	3473529	3473529 ▲	0	3473386	3473496 ▽	131
pxp_cz_2022_n162	15	6076090	6076090	0	6076090	6076090	0
pxp_de_2022_n165	15	5833705	5833280 ▽	0	5833705	5833705 ▲	4
pxp_dk_2022_n141	15	4795237	4795237 ▲	0	4795235	4795235 ▽	2
pxp_ee_2022_n126	15	4271813	4271813	0	4271813	4271813	0
pxp_es_2022_n162	15	5505970	5505970	0	5505909	5505970	61
pxp_fi_2022_n153	15	5823310	5823310	1	5823309	5823310	0
pxp_fr_2022_n164	15	6191534	6191534 ▲	31	6191529	6191503 ▽	36
pxp_gb_2022_n159	15	5322836	5322836 ▲	0	5322742	5322692 ▽	50
pxp_gr_2022_n130	15	3400458	3400458	0	3400458	3400458	0
pxp_hr_2022_n140	15	4686061	4686061	0	4686061	4686061	0
pxp_hu_2022_n160	15	5739389	5739235 ▽	4	5739389	5739389 ▲	3
pxp_id_2022_n132	15	5249010	5249010 ▲	0	5248855	5248495 ▽	528
pxp_ie_2022_n136	15	2351111	2351111	0	2351111	2351111	0
pxp_in_2022_n145	15	5026649	5026649 ▲	0	5026629	5026629 ▽	5
pxp_it_2022_n165	15	5986644	5986055 ▲	303	5986037	5986020 ▽	174
pxp_jp_2022_n159	15	6439663	6439663 ▲	0	6439659	6439659 ▽	4
pxp_kr_2022_n149	15	5254433	5254433 ▲	0	5254433	5253929 ▽	470
pxp_lt_2022_n133	15	3695421	3695421	0	3695421	3695421	0
pxp_lu_2022_n120	15	2534779	2534779 ▲	0	2534779	2534748 ▽	78
pxp_lv_2022_n132	15	4580756	4580756	0	4580756	4580756	0
pxp_mt_2022_n89	15	2345841	2345841	0	2345841	2345841	0
pxp_mx_2022_n139	15	4176737	4176712 ▲	270	4176446	4176446 ▽	0
pxp_nl_2022_n158	15	4248155	4248155	0	4248155	4248155	76
pxp_no_2022_n145	15	6631865	6631865 ▲	0	6631865	6631827 ▽	13
pxp_pl_2022_n166	15	5465401	5465401	0	5465401	5465401	24
pxp_pt_2022_n149	15	4974639	4974639 ▲	0	4974624	4973856 ▽	585
pxp_ro_2022_n151	15	5449486	5449486 ▲	0	5448898	5448906 ▽	631
pxp_ru_2022_n136	15	5230190	5230190 ▲	0	5229836	5230056 ▽	724
pxp_se_2022_n156	15	5079306	5079262 ▽	44	5079306	5079306 ▲	0
pxp_si_2022_n129	15	4262916	4262916	0	4262916	4262916	0
pxp_sk_2022_n151	15	5016909	5016909	0	5016909	5016909	746
pxp_tr_2022_n145	15	5365578	5365578 ▲	0	5365349	5365433 ▽	177
pxp_tw_2022_n138	15	5532757	5532757 ▲	0	5532672	5532556 ▽	156
pxp_us_2022_n170	15	6613891	6613891	0	6613891	6613891	0
pxp_wa_2022_n176	15	4011427	4011427	0	4011427	4011427	0
pxp_we_2022_n144	15	4492970	4492970 ▲	0	4492856	4492856 ▽	28
pxp_wf_2022_n141	15	4829905	4829905	0	4829905	4829905	0
pxp_wl_2022_n164	15	4966516	4966516 ▲	0	4966480	4966480 ▽	28
pxp_wm_2022_n161	15	3485302	3485302 ▲	0	3485251	3485292 ▽	47
pxp_za_2022_n134	15	5634755	5634755 ▲	0	5634755	5634406 ▽	799

Table 10: Results on MS-LOP for Φ metric and $m = 5$ on RXR instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
rxr_1995_n49	5	305714	305714	0	305714	305714	0
rxr_1996_n49	5	321398	321395 ∇	0	321398	321397 $\blacktriangle$	3
rxr_1997_n49	5	344171	344169 ∇	7	344171	344171 $\blacktriangle$	0
rxr_1998_n49	5	341353	341353	0	341353	341353	0
rxr_1999_n49	5	339578	339578	0	339578	339578	0
rxr_2000_n49	5	366789	366789	0	366789	366789	0
rxr_2001_n49	5	366136	366136	0	366136	366136	0
rxr_2002_n49	5	359941	359941	0	359941	359941	0
rxr_2003_n49	5	342297	342273	0	342273	342273	0
rxr_2004_n49	5	361730	361730	0	361730	361730	0
rxr_2005_n49	5	374472	374472	1	374472	374472	0
rxr_2006_n49	5	389869	389861 ∇	3	389869	389869 $\blacktriangle$	0
rxr_2007_n49	5	384025	384010	0	384010	384010	0
rxr_2008_n49	5	393623	393623	0	393623	393623	0
rxr_2009_n49	5	353102	353102	0	353102	353102	0
rxr_2010_n49	5	409574	409574	0	409574	409574	0
rxr_2011_n49	5	427121	427117	5	427117	427117	0
rxr_2012_n49	5	444297	444246	0	444246	444246	0
rxr_2013_n49	5	441715	441715	0	441715	441715	0
rxr_2014_n49	5	439131	439086 ∇	5	439131	439131 $\blacktriangle$	0
rxr_2015_n49	5	443533	443533	0	443533	443533	0
rxr_2016_n49	5	445787	445763 ∇	6	445787	445787 $\blacktriangle$	0
rxr_2017_n49	5	443845	443610	28	443630	443596	108
rxr_2018_n49	5	462885	462885	0	462885	462885	0
rxr_2019_n49	5	465456	465258 ∇	80	465456	465456 $\blacktriangle$	0
rxr_2020_n49	5	430099	430099	0	430099	430099	0
rxr_2021_n49	5	431894	431880 $\blacktriangle$	28	431866	431866 ∇	1
rxr_2022_n49	5	450772	450772	0	450772	450772	0

Table 11: Results on MS-LOP for Φ metric and $m = 10$ on RXR instances

Instance	m	MS-CD-RVNS			MS-MA-EDM				
		Best	Median	IQR	Best	Median	IQR		
rxr_1995_n49	10	305714	305625	▽	3	305629	305629	▲	0
rxr_1996_n49	10	321398	321395		21	321375	321375		7
rxr_1997_n49	10	344171	344146	▽	13	344152	344152	▲	1
rxr_1998_n49	10	341353	341283	▽	21	341353	341337	▲	16
rxr_1999_n49	10	339578	339550	▽	4	339557	339557	▲	0
rxr_2000_n49	10	366789	366788		0	366788	366788		0
rxr_2001_n49	10	366136	366084		0	366084	366084		5
rxr_2002_n49	10	359941	359927		0	359921	359927		1
rxr_2003_n49	10	342297	342211	▽	5	342253	342249	▲	18
rxr_2004_n49	10	361730	361691		0	361691	361691		0
rxr_2005_n49	10	374472	374439		7	374439	374429		31
rxr_2006_n49	10	389869	389824		17	389825	389828		16
rxr_2007_n49	10	384025	384001	▽	22	384001	384010	▲	10
rxr_2008_n49	10	393623	393585	▽	2	393588	393589	▲	4
rxr_2009_n49	10	353102	353082	▽	5	353085	353085	▲	0
rxr_2010_n49	10	409574	409394	▽	91	409536	409518	▲	18
rxr_2011_n49	10	427121	427117	▲	5	427082	427082	▽	0
rxr_2012_n49	10	444297	444212		26	444246	444210		26
rxr_2013_n49	10	441715	441564		0	441588	441564		33
rxr_2014_n49	10	439131	438783		127	438846	438844		82
rxr_2015_n49	10	443533	443364		9	443274	443369		90
rxr_2016_n49	10	445787	445580	▽	56	445742	445628	▲	116
rxr_2017_n49	10	443845	443377	▽	65	443630	443563	▲	115
rxr_2018_n49	10	462885	462799	▲	1	462797	462795	▽	27
rxr_2019_n49	10	465456	465258	▽	80	465456	465456	▲	0
rxr_2020_n49	10	430099	430090	▽	3	430094	430094	▲	1
rxr_2021_n49	10	431894	431880	▲	28	431846	431856	▽	19
rxr_2022_n49	10	450772	450753		0	450744	450753		1

Table 12: Results on MS-LOP for Φ metric and $m = 15$ on RXR instances

Instance	m	MS-CD-RVNS				MS-MA-EDM			
		Best	Median	IQR		Best	Median	IQR	
rxr_1995_n49	15	305714	305573	▽	4	305580	305580	▲	0
rxr_1996_n49	15	321398	321395	▲	21	321375	321375	▽	7
rxr_1997_n49	15	344171	344130	▽	15	344139	344138	▲	2
rxr_1998_n49	15	341353	341283	▽	21	341353	341337	▲	16
rxr_1999_n49	15	339578	339531	▽	3	339536	339536	▲	1
rxr_2000_n49	15	366789	366787		0	366787	366787		0
rxr_2001_n49	15	366136	366039	▽	0	366084	366054	▲	17
rxr_2002_n49	15	359941	359900	▲	9	359876	359892	▽	16
rxr_2003_n49	15	342297	342211	▽	5	342253	342249	▲	18
rxr_2004_n49	15	361730	361658	▽	5	361659	361660	▲	6
rxr_2005_n49	15	374472	374326	▽	14	374381	374361	▲	34
rxr_2006_n49	15	389869	389824		17	389825	389828		16
rxr_2007_n49	15	384025	384001	▽	22	384001	384010	▲	10
rxr_2008_n49	15	393623	393544	▽	7	393555	393553	▲	8
rxr_2009_n49	15	353102	353065	▽	14	353069	353069	▲	0
rxr_2010_n49	15	409574	409394	▽	101	409536	409518	▲	18
rxr_2011_n49	15	427121	427117	▲	5	427082	427082	▽	0
rxr_2012_n49	15	444297	444212		26	444246	444210		26
rxr_2013_n49	15	441715	441387	▽	114	441588	441560	▲	52
rxr_2014_n49	15	439131	438783		127	438773	438762		84
rxr_2015_n49	15	443533	443280	▽	132	443274	443369	▲	152
rxr_2016_n49	15	445787	445580	▲	53	445573	445494	▽	77
rxr_2017_n49	15	443845	443373	▽	64	443630	443563	▲	115
rxr_2018_n49	15	462885	462692	▽	23	462740	462736	▲	42
rxr_2019_n49	15	465456	465258	▽	80	465456	465456	▲	0
rxr_2020_n49	15	430099	430081	▽	3	430088	430088	▲	2
rxr_2021_n49	15	431894	431880	▲	28	431846	431856	▽	19
rxr_2022_n49	15	450772	450698	▽	16	450744	450738	▲	13

Table 13: Results on MS-LOP for Δ_{NN} metric and $m = 5$ on os300 instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
os300_at_2022_n300	5	8	25.0	34.75	7	17.0	15.75
os300_au_2022_n300	5	10	13.0	32.00	31	12.5	21.50
os300_be_2022_n300	5	11	14.5	15.50	11	27.0	27.00
os300_bg_2022_n300	5	30	32.0	28.00	42	30.5	41.00
os300_br_2022_n300	5	17	21.5	31.00	34	16.0	13.75
os300_ca_2022_n300	5	72	13.0	35.25	8	20.5	26.00
os300_ch_2022_n300	5	10	21.0	20.25	54	21.5	26.50
os300_cn_2022_n300	5	21	14.5	14.75	14	18.0	20.75
os300_cy_2022_n300	5	24	22.0	9.75	6	18.5	30.75
os300_cz_2022_n300	5	48	21.0	31.25	68	28.5	33.50
os300_de_2022_n300	5	7	25.0	30.50	37	26.5	25.75
os300_dk_2022_n300	5	15	26.0	39.50	50	19.5	22.00
os300_ee_2022_n300	5	10	18.0	27.75	26	32.0	21.50
os300_es_2022_n300	5	9	24.0	20.75	15	18.0	22.75
os300_fi_2022_n300	5	6	23.5	18.50	85	21.5	24.25
os300_fr_2022_n300	5	8	31.5	29.75	14	30.0	23.00
os300_gb_2022_n300	5	70	24.5	30.25	80	27.5	39.00
os300_gr_2022_n300	5	125	13.0	15.50	38	19.0	28.75
os300_hr_2022_n300	5	22	20.5	29.25	25	23.5	15.25
os300_hu_2022_n300	5	16	20.0	19.75	9	23.0	21.50
os300_id_2022_n300	5	18	20.0	22.00	24	26.0	23.25
os300_ie_2022_n300	5	7	9.5	17.75	6	12.0	13.75
os300_in_2022_n300	5	5	19.0	16.00	10	13.0	11.75
os300_it_2022_n300	5	75	28.5	37.75	21	21.0	29.00
os300_jp_2022_n300	5	37	20.5	25.25	43	27.0	30.75
os300_kr_2022_n300	5	79	13.0	19.50	9	13.5	30.75
os300_lt_2022_n300	5	47	21.0	34.75	165	28.5	35.25
os300_lu_2022_n300	5	34	10.5	26.00	101	17.5	22.00
os300_lv_2022_n300	5	59	27.0	22.25	46	26.5	23.75
os300_mt_2022_n300	5	127	28.0	29.75	11	14.5	10.25
os300_mx_2022_n300	5	15	16.0	18.50	80	20.0	19.75
os300_nl_2022_n300	5	32	19.0	17.75	18	22.5	26.50
os300_no_2022_n300	5	11	17.5	23.00	32	19.0	21.50
os300_pl_2022_n300	5	17	21.5	19.00	15	27.5	30.50
os300_pt_2022_n300	5	5	20.5	22.50	26	23.5	16.50
os300_ro_2022_n300	5	18	17.0	23.25	19	23.0	26.50
os300_ru_2022_n300	5	21	21.5	31.00	8	22.5	25.75
os300_se_2022_n300	5	55	33.0	42.25	16	28.5	23.00
os300_si_2022_n300	5	51	29.0	39.25	89	25.0	35.50
os300_sk_2022_n300	5	8	32.0	30.50	34	21.0	27.75
os300_tr_2022_n300	5	7	28.5	37.00	13	22.0	33.25
os300_tw_2022_n300	5	13	21.5	17.50	5	24.5	32.00
os300_us_2022_n300	5	33	17.5	14.75	8	18.0	15.25
os300_wa_2022_n300	5	59	23.0	46.75	33	24.0	20.25
os300_we_2022_n300	5	30	22.0	27.00	39	26.0	19.50
os300_wf_2022_n300	5	23	36.0	30.50	58	27.5	29.00
os300_wl_2022_n300	5	11	19.0	26.50	93	18.5	24.50
os300_wm_2022_n300	5	9	29.0	40.75	9	14.5	25.75
os300_za_2022_n300	5	10	14.5	27.25	5	19.5	25.50

Table 14: Results on MS-LOP for Δ_{NN} metric and $m = 10$ on os300 instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
os300_at_2022_n300	10	26	47.5 ▲	50.50	14	23.0 ▽	30.50
os300_au_2022_n300	10	10	34.0	52.50	104	21.0	49.25
os300_be_2022_n300	10	56	32.5	45.50	12	45.0	36.50
os300_bg_2022_n300	10	53	69.5 ▲	45.50	75	47.0 ▽	47.00
os300_br_2022_n300	10	87	53.5 ▲	49.75	69	32.0 ▽	37.75
os300_ca_2022_n300	10	70	50.5	56.75	12	25.5	40.50
os300_ch_2022_n300	10	15	44.0	35.25	55	49.5	39.75
os300_cn_2022_n300	10	102	42.5	52.25	19	28.5	31.25
os300_cy_2022_n300	10	157	59.5 ▲	90.00	14	31.5 ▽	51.25
os300_cz_2022_n300	10	51	51.0	44.00	53	41.0	39.75
os300_de_2022_n300	10	19	44.5	52.75	78	41.0	44.00
os300_dk_2022_n300	10	28	44.0	65.25	77	33.0	43.00
os300_ee_2022_n300	10	13	48.5	55.50	41	53.0	34.75
os300_es_2022_n300	10	21	47.0 ▲	43.25	10	27.0 ▽	29.50
os300_fi_2022_n300	10	13	56.5	57.25	69	38.5	49.25
os300_fr_2022_n300	10	62	70.5	42.25	29	42.0	40.75
os300_gb_2022_n300	10	77	59.5	70.00	88	43.0	55.75
os300_gr_2022_n300	10	162	31.5	51.50	144	44.0	47.00
os300_hr_2022_n300	10	44	53.0	43.50	50	31.5	34.00
os300_hu_2022_n300	10	30	43.0	38.75	29	42.0	36.00
os300_id_2022_n300	10	46	59.0	65.50	10	31.0	37.75
os300_ie_2022_n300	10	18	27.5	25.00	12	17.5	25.25
os300_in_2022_n300	10	19	40.0 ▲	18.25	17	20.0 ▽	31.75
os300_it_2022_n300	10	83	53.0	38.50	52	38.0	47.25
os300_jp_2022_n300	10	32	45.5	46.75	74	43.5	38.25
os300_kr_2022_n300	10	109	33.0	54.25	10	21.5	39.00
os300_lt_2022_n300	10	18	51.0	67.25	129	44.0	39.25
os300_lu_2022_n300	10	89	29.5	28.75	123	25.5	61.00
os300_lv_2022_n300	10	131	66.0	31.75	131	38.5	64.50
os300_mt_2022_n300	10	118	57.0 ▲	43.00	27	21.5 ▽	28.00
os300_mx_2022_n300	10	30	40.0	30.25	97	28.5	34.75
os300_nl_2022_n300	10	32	30.5	37.50	74	33.0	47.75
os300_no_2022_n300	10	52	48.5 ▲	43.00	44	16.5 ▽	31.00
os300_pl_2022_n300	10	37	47.5	49.75	44	44.0	41.75
os300_pt_2022_n300	10	21	58.5 ▲	74.00	59	35.0 ▽	34.50
os300_ro_2022_n300	10	51	50.5	52.25	10	51.0	69.00
os300_ru_2022_n300	10	41	42.5	52.50	28	38.5	40.50
os300_se_2022_n300	10	85	63.0	52.00	32	51.5	33.75
os300_si_2022_n300	10	101	70.0	56.00	83	49.5	27.25
os300_sk_2022_n300	10	14	55.0	62.50	58	40.0	39.50
os300_tr_2022_n300	10	98	77.5	70.50	31	57.0	56.75
os300_tw_2022_n300	10	25	41.5	46.00	15	34.5	38.25
os300_us_2022_n300	10	266	33.0	66.00	10	39.5	51.00
os300_wa_2022_n300	10	140	47.5	38.50	16	43.5	37.50
os300_we_2022_n300	10	60	45.5	46.75	52	55.0	29.50
os300_wf_2022_n300	10	43	67.0 ▲	55.50	42	41.5 ▽	50.75
os300_wl_2022_n300	10	52	45.0	45.25	47	31.5	29.75
os300_wm_2022_n300	10	20	42.5	67.75	224	35.5	46.25
os300_za_2022_n300	10	162	38.0	64.25	14	44.5	50.75

Table 15: Results on MS-LOP for Δ_{NN} metric and $m = 15$ on os300 instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
os300_at_2022_n300	15	49	81.0 ▲	71.75	19	35.5 ▽	46.50
os300_au_2022_n300	15	15	75.0	97.75	79	40.5	64.50
os300_be_2022_n300	15	82	58.0	49.00	18	63.0	54.75
os300_bg_2022_n300	15	78	93.0 ▲	58.50	90	64.5 ▽	58.75
os300_br_2022_n300	15	112	90.0 ▲	80.50	146	43.5 ▽	44.50
os300_ca_2022_n300	15	83	97.0 ▲	88.50	23	41.0 ▽	50.00
os300_ch_2022_n300	15	31	77.5	82.50	70	59.0	53.75
os300_cn_2022_n300	15	124	59.5	50.00	18	50.0	39.00
os300_cy_2022_n300	15	211	83.0 ▲	39.00	15	49.0 ▽	82.75
os300_cz_2022_n300	15	87	76.5	56.50	53	53.5	60.00
os300_de_2022_n300	15	29	70.5	73.00	194	65.0	66.50
os300_dk_2022_n300	15	39	60.0	69.25	93	52.5	57.75
os300_ee_2022_n300	15	16	78.5	51.50	86	79.0	44.00
os300_es_2022_n300	15	75	74.5 ▲	86.50	25	42.5 ▽	49.50
os300_fi_2022_n300	15	18	83.5	74.75	77	57.5	60.75
os300_fr_2022_n300	15	104	94.5	71.00	47	64.0	50.00
os300_gb_2022_n300	15	93	93.0	56.25	107	57.5	78.00
os300_gr_2022_n300	15	195	65.0	77.25	64	59.5	62.00
os300_hr_2022_n300	15	62	82.0 ▲	65.75	75	49.0 ▽	57.75
os300_hu_2022_n300	15	69	71.0	73.00	84	73.5	34.25
os300_id_2022_n300	15	65	87.5 ▲	98.50	22	53.0 ▽	58.50
os300_ie_2022_n300	15	41	41.0	33.50	25	25.0	35.75
os300_in_2022_n300	15	38	80.0 ▲	50.75	22	29.0 ▽	54.75
os300_it_2022_n300	15	151	89.5 ▲	58.50	228	53.5 ▽	66.25
os300_jp_2022_n300	15	558	72.5	78.25	101	66.5	61.50
os300_kr_2022_n300	15	127	57.0 ▲	76.75	19	33.0 ▽	50.00
os300_lt_2022_n300	15	116	89.0	59.50	136	77.0	53.25
os300_lu_2022_n300	15	105	44.5	42.50	109	27.0	57.25
os300_lv_2022_n300	15	121	99.5 ▲	65.25	110	60.0 ▽	49.00
os300_mt_2022_n300	15	184	82.5 ▲	44.75	30	31.5 ▽	39.25
os300_mx_2022_n300	15	113	68.5 ▲	57.25	113	43.5 ▽	52.75
os300_nl_2022_n300	15	42	59.5	58.00	52	48.0	43.25
os300_no_2022_n300	15	93	80.0 ▲	53.00	81	35.5 ▽	35.25
os300_pl_2022_n300	15	115	91.5 ▲	74.25	42	57.5 ▽	55.25
os300_pt_2022_n300	15	31	88.5 ▲	98.75	78	58.0 ▽	57.25
os300_ro_2022_n300	15	72	72.0	44.25	63	63.5	56.75
os300_ru_2022_n300	15	52	65.5	71.00	48	60.0	43.25
os300_se_2022_n300	15	105	87.5	64.75	64	79.0	48.00
os300_si_2022_n300	15	113	84.0	82.75	123	67.0	49.75
os300_sk_2022_n300	15	20	77.5	71.25	91	58.5	59.25
os300_tr_2022_n300	15	103	109.0	72.00	41	68.5	95.50
os300_tw_2022_n300	15	39	65.0	58.25	20	61.0	49.25
os300_us_2022_n300	15	71	67.5	71.25	18	53.5	56.50
os300_wa_2022_n300	15	131	77.5	76.00	17	68.0	49.50
os300_we_2022_n300	15	100	88.5	64.25	53	70.5	33.00
os300_wf_2022_n300	15	86	96.0 ▲	61.75	42	54.0 ▽	64.75
os300_wl_2022_n300	15	66	69.0 ▲	66.25	64	44.0 ▽	47.00
os300_wm_2022_n300	15	31	92.0 ▲	86.75	120	41.5 ▽	59.75
os300_za_2022_n300	15	197	62.5	80.00	15	53.0	67.50

Table 16: Results on MS-LOP for Δ_{NN} metric and $m = 5$ on PXP instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
pxp_at_2022_n155	5	206	207.0 ▽	32.00	308	300.0 ▲	33.50
pxp_au_2022_n155	5	232	291.0 ▽	85.00	410	427.5 ▲	64.75
pxp_be_2022_n155	5	413	395.5 ▽	68.50	472	539.5 ▲	74.75
pxp_bg_2022_n135	5	266	219.0 ▽	81.00	333	323.0 ▲	28.50
pxp_br_2022_n155	5	170	193.5	36.00	208	193.0	32.25
pxp_ca_2022_n158	5	559	571.5 ▽	27.00	797	757.0 ▲	61.75
pxp_ch_2022_n141	5	433	398.0 ▽	94.25	745	743.0 ▲	35.50
pxp_cn_2022_n169	5	496	482.5	75.75	652	479.0	133.75
pxp_cy_2022_n116	5	108	109.0 ▽	12.75	170	178.5 ▲	34.75
pxp_cz_2022_n162	5	473	383.5 ▽	98.75	743	722.0 ▲	72.00
pxp_de_2022_n165	5	255	234.5 ▽	65.50	478	485.0 ▲	92.75
pxp_dk_2022_n141	5	131	153.0 ▽	26.50	200	179.5 ▲	50.75
pxp_ee_2022_n126	5	257	239.5 ▽	84.00	452	450.5 ▲	30.25
pxp_es_2022_n162	5	193	252.5 ▽	77.50	412	401.5 ▲	53.75
pxp_fi_2022_n153	5	199	181.5 ▽	29.75	196	215.5 ▲	36.75
pxp_fr_2022_n164	5	636	531.0 ▽	163.25	671	671.5 ▲	148.50
pxp_gb_2022_n159	5	690	720.0 ▽	60.25	703	890.0 ▲	197.25
pxp_gr_2022_n130	5	772	771.0 ▽	146.00	1185	1172.5 ▲	102.75
pxp_hr_2022_n140	5	596	568.5 ▽	106.25	956	929.5 ▲	90.75
pxp_hu_2022_n160	5	207	189.5 ▽	61.25	261	294.0 ▲	60.00
pxp_id_2022_n132	5	261	262.0 ▲	28.75	181	239.0 ▽	55.00
pxp_ie_2022_n136	5	779	904.0 ▽	140.25	1098	1167.0 ▲	163.25
pxp_in_2022_n145	5	278	260.0 ▽	33.75	325	313.0 ▲	57.75
pxp_it_2022_n165	5	190	271.5 ▽	57.25	291	349.5 ▲	69.50
pxp_jp_2022_n159	5	156	119.5 ▽	30.50	217	188.5 ▲	26.50
pxp_kr_2022_n149	5	251	222.0	38.00	214	201.5	45.75
pxp_lt_2022_n133	5	275	366.0 ▽	65.25	613	540.0 ▲	62.75
pxp_lu_2022_n120	5	798	743.5 ▽	46.75	838	812.5 ▲	101.25
pxp_lv_2022_n132	5	291	258.5 ▽	58.75	407	415.5 ▲	37.75
pxp_mt_2022_n89	5	404	369.0 ▽	27.75	402	424.0 ▲	59.75
pxp_mx_2022_n139	5	158	157.5 ▽	28.50	307	286.0 ▲	32.75
pxp_nl_2022_n158	5	585	569.0 ▽	138.75	659	707.5 ▲	94.25
pxp_no_2022_n145	5	321	378.5 ▽	61.00	512	561.5 ▲	169.00
pxp_pl_2022_n166	5	182	223.0 ▽	57.50	291	319.0 ▲	31.25
pxp_pt_2022_n149	5	197	196.5 ▽	45.50	217	234.5 ▲	58.50
pxp_ro_2022_n151	5	215	344.0 ▽	38.50	662	466.0 ▲	83.00
pxp_ru_2022_n136	5	92	104.0 ▽	15.75	161	157.5 ▲	30.50
pxp_se_2022_n156	5	281	279.5 ▽	38.50	380	409.5 ▲	50.50
pxp_si_2022_n129	5	237	219.5 ▽	34.75	162	257.5 ▲	56.75
pxp_sk_2022_n151	5	238	219.5	35.50	190	213.5	48.00
pxp_tr_2022_n145	5	188	187.5 ▲	91.50	90	144.5 ▽	29.00
pxp_tw_2022_n138	5	470	391.5 ▽	45.00	425	478.0 ▲	88.25
pxp_us_2022_n170	5	320	326.5 ▽	57.50	507	495.0 ▲	42.50
pxp_wa_2022_n176	5	233	190.5 ▽	33.75	326	340.5 ▲	45.50
pxp_we_2022_n144	5	42	41.0 ▽	9.50	70	62.0 ▲	19.25
pxp_wf_2022_n141	5	111	90.0 ▽	16.00	122	112.5 ▲	17.50
pxp_wl_2022_n164	5	56	72.0 ▽	13.75	133	127.0 ▲	16.25
pxp_wm_2022_n161	5	355	274.0 ▽	59.75	500	518.0 ▲	33.75
pxp_za_2022_n134	5	164	154.0	18.50	160	170.5	43.25

Table 17: Results on MS-LOP for Δ_{NN} metric and $m = 10$ on PXP instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
pxp_at_2022_n155	10	317	313.5 ▽	51.00	537	504.0 ▲	70.75
pxp_au_2022_n155	10	349	457.5 ▽	159.00	677	728.0 ▲	114.75
pxp_be_2022_n155	10	630	605.5 ▽	127.75	767	909.0 ▲	146.50
pxp_bg_2022_n135	10	391	324.5 ▽	123.25	560	517.0 ▲	51.25
pxp_br_2022_n155	10	260	295.0 ▽	41.00	340	315.5 ▲	58.50
pxp_ca_2022_n158	10	907	955.5 ▽	81.75	1348	1338.5 ▲	99.75
pxp_ch_2022_n141	10	574	583.5 ▽	132.50	1252	1244.5 ▲	62.25
pxp_cn_2022_n169	10	685	712.5 ▽	82.25	1047	791.5 ▲	219.00
pxp_cy_2022_n116	10	161	163.0 ▽	19.50	280	291.5 ▲	66.75
pxp_cz_2022_n162	10	714	590.0 ▽	147.75	1285	1271.0 ▲	126.75
pxp_de_2022_n165	10	389	363.0 ▽	115.00	811	823.0 ▲	176.00
pxp_dk_2022_n141	10	200	224.0 ▽	56.75	331	292.0 ▲	79.50
pxp_ee_2022_n126	10	381	338.5 ▽	127.25	756	757.0 ▲	67.25
pxp_es_2022_n162	10	277	354.0 ▽	123.50	715	666.0 ▲	113.25
pxp_fi_2022_n153	10	274	265.5 ▽	48.50	331	354.0 ▲	47.50
pxp_fr_2022_n164	10	998	774.0 ▽	280.00	1084	1162.5 ▲	241.00
pxp_gb_2022_n159	10	1100	1095.0 ▽	188.00	1249	1541.0 ▲	293.25
pxp_gr_2022_n130	10	1079	1146.0 ▽	222.50	2051	2072.0 ▲	152.00
pxp_hr_2022_n140	10	877	895.0 ▽	175.50	1696	1642.0 ▲	177.00
pxp_hu_2022_n160	10	335	283.5 ▽	99.00	419	480.0 ▲	120.00
pxp_id_2022_n132	10	406	398.0	57.75	285	396.0	81.25
pxp_ie_2022_n136	10	1259	1419.0 ▽	221.75	1806	2049.0 ▲	311.25
pxp_in_2022_n145	10	423	400.5 ▽	55.75	528	517.5 ▲	90.25
pxp_it_2022_n165	10	260	389.0 ▽	108.00	501	569.0 ▲	129.75
pxp_jp_2022_n159	10	231	183.5 ▽	42.75	362	318.0 ▲	54.75
pxp_kr_2022_n149	10	352	316.5	41.50	339	303.0	68.50
pxp_lt_2022_n133	10	394	544.0 ▽	122.00	1028	895.0 ▲	96.75
pxp_lu_2022_n120	10	1187	1155.0 ▽	118.50	1417	1361.5 ▲	135.50
pxp_lv_2022_n132	10	450	379.0 ▽	94.00	698	701.5 ▲	58.75
pxp_mt_2022_n89	10	605	560.0 ▽	65.75	656	682.0 ▲	111.75
pxp_mx_2022_n139	10	243	233.5 ▽	42.00	505	471.0 ▲	61.25
pxp_nl_2022_n158	10	939	823.0 ▽	233.50	1143	1230.0 ▲	154.00
pxp_no_2022_n145	10	469	573.0 ▽	92.00	888	935.5 ▲	273.25
pxp_pl_2022_n166	10	253	360.5 ▽	69.75	464	532.5 ▲	53.50
pxp_pt_2022_n149	10	280	282.0 ▽	75.00	321	375.5 ▲	90.25
pxp_ro_2022_n151	10	308	545.5 ▽	75.50	1099	773.5 ▲	150.25
pxp_ru_2022_n136	10	149	150.5 ▽	32.50	250	251.5 ▲	66.25
pxp_se_2022_n156	10	422	416.5 ▽	76.00	654	662.5 ▲	81.50
pxp_si_2022_n129	10	363	337.5 ▽	73.00	260	409.0 ▲	91.25
pxp_sk_2022_n151	10	372	318.0	68.50	293	330.0	78.50
pxp_tr_2022_n145	10	220	253.0	89.25	134	224.0	56.00
pxp_tw_2022_n138	10	780	606.5 ▽	85.25	728	828.5 ▲	156.75
pxp_us_2022_n170	10	521	544.5 ▽	81.25	873	868.5 ▲	73.75
pxp_wa_2022_n176	10	390	291.0 ▽	79.25	564	568.5 ▲	86.50
pxp_we_2022_n144	10	54	54.0 ▽	18.25	102	91.5 ▲	33.50
pxp_wf_2022_n141	10	182	138.0 ▽	26.50	189	178.5 ▲	25.00
pxp_wl_2022_n164	10	88	102.0 ▽	15.75	219	206.5 ▲	31.00
pxp_wm_2022_n161	10	557	423.0 ▽	95.00	862	889.0 ▲	51.75
pxp_za_2022_n134	10	264	241.5 ▽	33.00	274	283.5 ▲	72.50

Table 18: Results on MS-LOP for Δ_{NN} metric and $m = 15$ on PXP instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
pxp_at_2022_n155	15	438	404.0 ▽	98.00	730	692.0 ▲	87.00
pxp_au_2022_n155	15	473	576.0 ▽	223.75	922	1010.0 ▲	169.50
pxp_be_2022_n155	15	797	756.5 ▽	161.25	1060	1259.0 ▲	209.25
pxp_bg_2022_n135	15	484	399.0 ▽	185.50	750	705.0 ▲	72.75
pxp_br_2022_n155	15	333	378.0 ▽	54.50	462	432.0 ▲	87.00
pxp_ca_2022_n158	15	1186	1252.5 ▽	166.25	1916	1874.0 ▲	151.75
pxp_ch_2022_n141	15	678	718.0 ▽	191.25	1695	1703.0 ▲	77.75
pxp_cn_2022_n169	15	864	881.5 ▽	138.50	1424	1070.5 ▲	291.75
pxp_cy_2022_n116	15	212	206.0 ▽	23.00	376	395.5 ▲	100.75
pxp_cz_2022_n162	15	943	733.5 ▽	223.00	1796	1756.0 ▲	180.50
pxp_de_2022_n165	15	523	457.5 ▽	161.00	1112	1153.5 ▲	239.00
pxp_dk_2022_n141	15	240	281.5 ▽	81.75	452	395.0 ▲	113.25
pxp_ee_2022_n126	15	481	428.5 ▽	206.00	1025	1053.5 ▲	74.00
pxp_es_2022_n162	15	333	470.5 ▽	122.75	986	903.5 ▲	166.25
pxp_fi_2022_n153	15	365	347.5 ▽	64.00	462	488.0 ▲	59.25
pxp_fr_2022_n164	15	1350	1037.5 ▽	380.50	1530	1632.0 ▲	356.25
pxp_gb_2022_n159	15	1433	1390.0 ▽	194.00	1758	2107.0 ▲	397.75
pxp_gr_2022_n130	15	1396	1439.5 ▽	293.25	2904	2919.5 ▲	276.25
pxp_hr_2022_n140	15	1071	1178.5 ▽	285.00	2378	2311.5 ▲	233.00
pxp_hu_2022_n160	15	425	322.5 ▽	125.00	566	652.5 ▲	165.00
pxp_id_2022_n132	15	511	506.0 ▽	57.75	392	536.0 ▲	109.25
pxp_ie_2022_n136	15	1661	1874.0 ▽	282.50	2477	2873.0 ▲	409.00
pxp_in_2022_n145	15	569	502.0 ▽	64.50	700	694.0 ▲	135.25
pxp_it_2022_n165	15	323	477.0 ▽	145.50	700	779.0 ▲	164.25
pxp_jp_2022_n159	15	311	235.5 ▽	60.25	488	433.0 ▲	91.75
pxp_kr_2022_n149	15	444	387.5 ▽	78.75	449	398.5 ▽	102.00
pxp_lt_2022_n133	15	469	701.5 ▽	158.50	1448	1235.0 ▲	143.25
pxp_lu_2022_n120	15	1548	1540.0 ▽	183.00	1934	1895.5 ▲	201.00
pxp_lv_2022_n132	15	573	454.0 ▽	112.75	936	975.0 ▲	79.25
pxp_mt_2022_n89	15	758	746.5 ▽	77.25	865	926.5 ▲	152.25
pxp_mx_2022_n139	15	307	286.0 ▽	44.75	685	645.5 ▲	92.50
pxp_nl_2022_n158	15	1160	1107.5 ▽	324.50	1613	1726.5 ▲	217.25
pxp_no_2022_n145	15	581	746.5 ▽	130.00	1214	1295.5 ▲	370.50
pxp_pl_2022_n166	15	337	459.5 ▽	123.50	628	722.0 ▲	86.00
pxp_pt_2022_n149	15	368	348.5 ▽	86.25	411	499.5 ▲	134.00
pxp_ro_2022_n151	15	403	696.5 ▽	95.25	1490	1067.0 ▲	224.75
pxp_ru_2022_n136	15	190	183.5 ▽	50.75	325	332.0 ▲	90.00
pxp_se_2022_n156	15	558	549.5 ▽	108.25	887	904.0 ▲	105.75
pxp_si_2022_n129	15	496	421.0 ▽	86.75	342	548.5 ▲	115.00
pxp_sk_2022_n151	15	490	418.5 ▽	78.50	397	426.5 ▽	118.75
pxp_tr_2022_n145	15	264	275.0 ▽	114.25	169	285.0 ▽	71.00
pxp_tw_2022_n138	15	1066	783.0 ▽	119.50	988	1149.0 ▲	215.00
pxp_us_2022_n170	15	697	713.0 ▽	131.25	1230	1212.0 ▲	102.25
pxp_wa_2022_n176	15	524	374.5 ▽	113.00	764	790.5 ▲	129.25
pxp_we_2022_n144	15	66	61.5 ▽	25.25	132	121.5 ▲	46.50
pxp_wf_2022_n141	15	236	176.0 ▽	36.25	251	239.0 ▲	40.00
pxp_wl_2022_n164	15	114	124.5 ▽	19.25	295	280.5 ▲	41.25
pxp_wm_2022_n161	15	658	534.5 ▽	142.50	1195	1237.5 ▲	76.00
pxp_za_2022_n134	15	357	313.5 ▽	52.50	376	386.5 ▲	109.25

Table 19: Results on MS-LOP for Δ_{NN} metric and $m = 5$ on RXR instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
rxr_1995_n49	5	19	19.0	0.00	19	19.0	0.00
rxr_1996_n49	5	50	50.0 ∇	0.00	51	51.0 $\blacktriangle$	1.00
rxr_1997_n49	5	31	31.0 $\blacktriangle$	34.00	23	23.0 ∇	0.00
rxr_1998_n49	5	47	47.0	0.00	47	47.0	0.00
rxr_1999_n49	5	19	19.0	0.00	19	19.0	0.00
rxr_2000_n49	5	15	15.0	0.00	15	15.0	0.00
rxr_2001_n49	5	8	8.0	0.00	8	8.0	0.00
rxr_2002_n49	5	39	39.0	0.00	39	39.0	0.00
rxr_2003_n49	5	38	38.0	0.00	38	38.0	0.00
rxr_2004_n49	5	31	31.0	0.00	31	31.0	0.00
rxr_2005_n49	5	58	40.0	18.00	40	40.0	0.00
rxr_2006_n49	5	36	36.0 $\blacktriangle$	20.00	35	35.0 ∇	0.00
rxr_2007_n49	5	39	39.0	0.00	39	39.0	0.00
rxr_2008_n49	5	33	33.0	0.00	33	33.0	0.00
rxr_2009_n49	5	8	8.0	0.00	8	8.0	0.00
rxr_2010_n49	5	38	38.0	0.00	38	38.0	0.00
rxr_2011_n49	5	29	29.0	1.00	29	29.0	0.00
rxr_2012_n49	5	31	31.0	0.00	31	31.0	0.00
rxr_2013_n49	5	54	54.0	0.00	54	54.0	0.00
rxr_2014_n49	5	43	43.0 $\blacktriangle$	1.75	41	41.0 ∇	0.00
rxr_2015_n49	5	20	20.0	0.00	20	20.0	0.00
rxr_2016_n49	5	26	26.0 $\blacktriangle$	5.00	21	21.0 ∇	0.00
rxr_2017_n49	5	77	83.0	7.00	78	76.5	19.50
rxr_2018_n49	5	35	35.0	0.00	35	35.0	0.00
rxr_2019_n49	5	63	63.0 $\blacktriangle$	11.25	28	28.0 ∇	0.00
rxr_2020_n49	5	28	28.0	0.00	28	28.0	0.00
rxr_2021_n49	5	53	46.5 ∇	13.00	53	53.0 $\blacktriangle$	0.00
rxr_2022_n49	5	24	24.0	0.00	24	24.0	0.00

Table 20: Results on MS-LOP for Δ_{NN} metric and $m = 10$ on RXR instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
rxr_1995_n49	10	50	59.0 $\blacktriangle$	15.00	42	42.0 ∇	0.00
rxr_1996_n49	10	50	50.0 ∇	11.00	68	68.0 $\blacktriangle$	7.00
rxr_1997_n49	10	61	42.0 $\blacktriangle$	53.00	10	10.0 ∇	14.00
rxr_1998_n49	10	90	83.0 $\blacktriangle$	12.00	47	56.0 ∇	9.00
rxr_1999_n49	10	41	41.0 $\blacktriangle$	14.00	30	30.0 ∇	0.00
rxr_2000_n49	10	20	20.0	0.00	20	20.0	3.00
rxr_2001_n49	10	35	35.0	0.00	31	35.0	9.25
rxr_2002_n49	10	56	56.0	0.00	59	56.0	0.75
rxr_2003_n49	10	84	80.0 $\blacktriangle$	20.00	51	38.0 ∇	25.25
rxr_2004_n49	10	61	61.0	0.00	61	61.0	0.00
rxr_2005_n49	10	91	90.0 $\blacktriangle$	1.00	90	84.0 ∇	21.50
rxr_2006_n49	10	46	59.5 $\blacktriangle$	11.00	57	48.0 ∇	9.75
rxr_2007_n49	10	55	38.0	17.00	38	38.0	1.00
rxr_2008_n49	10	70	70.0 $\blacktriangle$	5.00	75	65.0 ∇	3.50
rxr_2009_n49	10	36	21.0 $\blacktriangle$	26.00	10	10.0 ∇	0.00
rxr_2010_n49	10	90	83.5 $\blacktriangle$	8.00	54	70.0 ∇	16.00
rxr_2011_n49	10	29	29.0 ∇	1.00	30	30.0 $\blacktriangle$	0.00
rxr_2012_n49	10	36	36.0 $\blacktriangle$	2.00	31	31.0 ∇	8.00
rxr_2013_n49	10	78	78.0	0.00	74	78.0	6.75
rxr_2014_n49	10	57	59.0 ∇	8.00	72	72.0 $\blacktriangle$	8.75
rxr_2015_n49	10	32	54.0	9.50	77	46.0	19.50
rxr_2016_n49	10	43	42.5 $\blacktriangle$	5.75	20	23.0 ∇	8.50
rxr_2017_n49	10	103	115.5 $\blacktriangle$	18.50	78	79.0 ∇	20.00
rxr_2018_n49	10	54	56.0	14.00	53	56.0	15.25
rxr_2019_n49	10	63	63.0 $\blacktriangle$	11.25	28	28.0 ∇	0.00
rxr_2020_n49	10	64	60.0 $\blacktriangle$	10.00	52	52.0 ∇	1.00
rxr_2021_n49	10	53	46.5 ∇	13.00	60	53.0 $\blacktriangle$	7.00
rxr_2022_n49	10	40	40.0	0.00	52	40.0	7.50

Table 21: Results on MS-LOP for Δ_{NN} metric and $m = 15$ on RXR instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
rxr_1995_n49	15	131	112.0 $\blacktriangle$	30.00	59	69.0 ∇	18.00
rxr_1996_n49	15	50	50.0 ∇	11.00	68	68.0 $\blacktriangle$	7.00
rxr_1997_n49	15	114	71.0 $\blacktriangle$	52.75	26	28.0 ∇	10.00
rxr_1998_n49	15	90	83.0 $\blacktriangle$	12.00	47	56.0 ∇	9.00
rxr_1999_n49	15	51	56.0	6.75	55	56.0	9.00
rxr_2000_n49	15	37	32.0	4.00	33	30.0	6.25
rxr_2001_n49	15	43	43.0 $\blacktriangle$	0.00	31	39.0 ∇	5.50
rxr_2002_n49	15	84	82.0 ∇	2.00	98	86.0 $\blacktriangle$	7.00
rxr_2003_n49	15	84	80.0 $\blacktriangle$	20.00	51	38.0 ∇	25.25
rxr_2004_n49	15	106	96.5	10.75	114	93.0	12.25
rxr_2005_n49	15	109	109.0 $\blacktriangle$	5.00	106	99.0 ∇	19.25
rxr_2006_n49	15	46	59.5 $\blacktriangle$	11.00	57	48.0 ∇	9.75
rxr_2007_n49	15	55	38.0	17.00	38	38.0	1.00
rxr_2008_n49	15	102	102.0	12.25	105	103.5	6.75
rxr_2009_n49	15	49	38.0 $\blacktriangle$	42.50	25	25.0 ∇	0.00
rxr_2010_n49	15	90	84.0 $\blacktriangle$	11.75	54	70.0 ∇	16.00
rxr_2011_n49	15	29	29.0 ∇	1.00	30	30.0 $\blacktriangle$	0.00
rxr_2012_n49	15	36	36.0 $\blacktriangle$	2.00	31	31.0 ∇	8.00
rxr_2013_n49	15	109	109.0 $\blacktriangle$	17.00	74	79.0 ∇	11.00
rxr_2014_n49	15	57	59.0 ∇	8.00	85	81.0 $\blacktriangle$	11.00
rxr_2015_n49	15	79	54.0	17.00	77	46.0	19.50
rxr_2016_n49	15	43	42.5 $\blacktriangle$	5.75	31	31.0 ∇	9.00
rxr_2017_n49	15	103	116.5 $\blacktriangle$	20.50	78	79.0 ∇	20.00
rxr_2018_n49	15	101	86.0 $\blacktriangle$	15.50	88	77.5 ∇	15.50
rxr_2019_n49	15	63	63.0 $\blacktriangle$	11.25	28	28.0 ∇	0.00
rxr_2020_n49	15	71	69.0 ∇	7.00	88	84.5 $\blacktriangle$	12.00
rxr_2021_n49	15	53	46.5 ∇	13.00	60	53.0 $\blacktriangle$	7.00
rxr_2022_n49	15	70	70.0 $\blacktriangle$	5.00	52	53.0 ∇	7.50

Table 22: Results on MS-LOP for Δ_{SP} metric and $m = 5$ on os300 instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
os300_at_2022_n300	5	4.4	4.52	0.58	4.26	4.64	0.59
os300_au_2022_n300	5	4.49	4.57	0.59	5.0	4.4	0.61
os300_be_2022_n300	5	4.68	4.68	0.59	3.89	4.97	0.60
os300_bg_2022_n300	5	5.0	4.99	0.31	5.0	4.98	0.60
os300_br_2022_n300	5	4.63	4.69	0.59	5.0	4.4	0.58
os300_ca_2022_n300	5	5.0	4.54	0.66	3.89	4.52	0.61
os300_ch_2022_n300	5	3.89	4.7	0.59	5.0	4.98	0.60
os300_cn_2022_n300	5	4.4	4.4	0.61	4.38	4.55	0.60
os300_cy_2022_n300	5	5.0	4.98	0.52	3.94	4.88	0.60
os300_cz_2022_n300	5	5.0	4.84	0.61	4.99	4.99	0.60
os300_de_2022_n300	5	4.38	4.72	0.60	5.0	4.98	0.60
os300_dk_2022_n300	5	4.4	4.4	0.64	5.0	4.54	0.60
os300_ee_2022_n300	5	4.34	4.69	0.60	4.98	5.0	0.03
os300_es_2022_n300	5	4.4	4.95	0.59	4.4	4.62	0.59
os300_fi_2022_n300	5	3.94	4.7	0.65	5.0	4.96	0.69
os300_fr_2022_n300	5	4.4	4.82	0.60	4.4	4.79	0.60
os300_gb_2022_n300	5	4.69	4.7	0.59	5.0	4.98	0.60
os300_gr_2022_n300	5	5.0	4.4	0.57	5.0	4.69	0.61
os300_hr_2022_n300	5	5.0	4.7	0.60	5.0	4.79	0.60
os300_hu_2022_n300	5	4.97	4.7	0.59	4.4	4.97	0.54
os300_id_2022_n300	5	4.7	4.7	0.58	3.89	4.64	0.60
os300_ie_2022_n300	5	4.34	4.36	0.81	3.94	4.2	0.87
os300_in_2022_n300	5	4.08	4.4	0.31	3.89	4.4	0.93
os300_it_2022_n300	5	5.0	4.93	0.60	5.0	4.64	0.60
os300_jp_2022_n300	5	4.97	4.7	0.61	5.0	4.78	0.60
os300_kr_2022_n300	5	4.4	4.54	0.48	4.38	4.4	0.61
os300_lt_2022_n300	5	4.69	4.69	0.60	5.0	4.99	0.60
os300_lu_2022_n300	5	4.4	4.38	0.60	5.0	4.4	0.61
os300_lv_2022_n300	5	5.0	4.98	0.60	5.0	4.98	0.60
os300_mt_2022_n300	5	5.0	4.69	0.43	3.89	4.4	0.97
os300_mx_2022_n300	5	4.95	4.63	0.57	5.0	4.79	0.60
os300_nl_2022_n300	5	4.4	4.4	0.55	4.98	4.98	0.60
os300_no_2022_n300	5	4.87	4.7	0.58	5.0	4.38	1.01
os300_pl_2022_n300	5	4.97	4.87	0.31	4.97	4.97	0.60
os300_pt_2022_n300	5	3.59	4.7	0.61	5.0	4.96	0.60
os300_ro_2022_n300	5	4.99	4.68	0.72	4.38	4.84	0.60
os300_ru_2022_n300	5	4.98	4.69	0.58	4.4	4.84	0.60
os300_se_2022_n300	5	5.0	4.84	0.60	4.98	4.99	0.46
os300_si_2022_n300	5	4.98	4.95	0.60	5.0	4.98	0.46
os300_sk_2022_n300	5	4.45	4.57	0.59	5.0	4.88	0.60
os300_tr_2022_n300	5	4.13	4.79	0.31	4.69	4.69	0.60
os300_tw_2022_n300	5	4.4	4.54	0.67	4.1	4.91	0.60
os300_us_2022_n300	5	4.7	4.69	0.46	4.39	4.55	0.68
os300_wa_2022_n300	5	5.0	4.7	0.60	4.38	4.84	0.60
os300_we_2022_n300	5	4.99	4.96	0.55	4.99	4.99	0.02
os300_wf_2022_n300	5	5.0	4.99	0.60	4.99	4.99	0.60
os300_wl_2022_n300	5	4.4	4.69	0.60	5.0	4.4	0.61
os300_wm_2022_n300	5	4.38	4.7	0.65	3.89	4.4	0.63
os300_za_2022_n300	5	4.4	4.4	0.30	4.08	4.78	0.60

Table 23: Results on MS-LOP for Δ_{SP} metric and $m = 10$ on os300 instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
os300_at_2022_n300	10	9.09	9.1	1.16	7.59	9.39	1.45
os300_au_2022_n300	10	6.25	8.8	1.73	10.0	8.79	1.51
os300_be_2022_n300	10	9.67	9.37	1.40	7.71	9.72	1.12
os300_bg_2022_n300	10	9.99	9.98	0.53	10.0	9.94	1.20
os300_br_2022_n300	10	9.6	9.5	0.89	10.0	8.8	1.47
os300_ca_2022_n300	10	10.0	9.23	1.41	7.71	8.93	1.50
os300_ch_2022_n300	10	7.99	9.4	0.91	9.99	9.95	1.20
os300_cn_2022_n300	10	9.09	9.08	1.15	8.76	9.4	1.46
os300_cy_2022_n300	10	9.99	9.96	0.60	6.9	9.72	1.28
os300_cz_2022_n300	10	9.97	9.54	1.13	9.96	9.81	1.23
os300_de_2022_n300	10	8.75	9.62	1.18	10.0	9.95	1.20
os300_dk_2022_n300	10	8.8	9.09	1.45	10.0	9.38	1.25
os300_ee_2022_n300	10	8.61	9.38	1.18	9.95	9.98	0.05
os300_es_2022_n300	10	8.8	9.77	0.88	8.5	9.21	1.20
os300_fi_2022_n300	10	7.39	9.4	1.42	10.0	9.93	1.30
os300_fr_2022_n300	10	9.1	9.77	0.92	8.8	9.73	1.19
os300_gb_2022_n300	10	9.38	9.4	0.82	10.0	9.92	1.42
os300_gr_2022_n300	10	9.98	8.88	1.17	9.97	9.38	1.40
os300_hr_2022_n300	10	9.99	9.33	0.99	10.0	9.55	1.43
os300_hu_2022_n300	10	9.91	9.66	0.87	9.39	9.91	1.12
os300_id_2022_n300	10	9.7	9.69	0.90	7.69	9.58	1.24
os300_ie_2022_n300	10	8.5	8.67	0.85	6.85	8.29	1.27
os300_in_2022_n300	10	8.46	9.19	0.75	7.99	8.8	1.75
os300_it_2022_n300	10	10.0	9.82	0.89	9.99	9.4	1.48
os300_jp_2022_n300	10	9.95	9.39	1.19	10.0	9.67	1.25
os300_kr_2022_n300	10	9.06	9.1	1.37	8.46	8.78	1.50
os300_lt_2022_n300	10	9.07	9.38	1.21	10.0	9.95	1.22
os300_lu_2022_n300	10	9.4	8.73	0.87	10.0	8.79	1.53
os300_lv_2022_n300	10	9.99	9.98	0.92	10.0	9.94	1.22
os300_mt_2022_n300	10	9.99	9.38	1.21	7.77	8.79	1.91
os300_mx_2022_n300	10	9.91	9.22	1.36	10.0	9.28	1.19
os300_nl_2022_n300	10	8.8	8.8	1.10	9.96	9.93	1.23
os300_no_2022_n300	10	9.79	9.39	1.05	9.99	8.64	1.94
os300_pl_2022_n300	10	9.95	9.74	0.83	9.64	9.84	1.24
os300_pt_2022_n300	10	7.99	9.68	1.24	10.0	9.89	1.22
os300_ro_2022_n300	10	9.94	9.1	1.05	8.46	9.8	1.50
os300_ru_2022_n300	10	9.96	9.24	1.36	8.79	9.66	1.42
os300_se_2022_n300	10	10.0	9.67	0.92	9.94	9.98	0.82
os300_si_2022_n300	10	9.96	9.85	0.90	9.99	9.97	0.70
os300_sk_2022_n300	10	8.11	9.1	1.16	10.0	9.76	1.20
os300_tr_2022_n300	10	8.11	9.65	0.64	9.38	9.24	1.21
os300_tw_2022_n300	10	9.09	9.38	1.04	7.99	9.78	1.24
os300_us_2022_n300	10	9.39	9.36	0.93	8.48	9.1	1.90
os300_wa_2022_n300	10	9.99	9.38	1.09	8.76	9.54	1.23
os300_we_2022_n300	10	9.98	9.91	0.68	9.97	9.98	0.04
os300_wf_2022_n300	10	9.99	9.89	0.65	9.98	9.96	1.24
os300_wl_2022_n300	10	9.09	9.39	1.35	9.98	8.8	1.23
os300_wm_2022_n300	10	9.06	9.37	1.33	7.78	8.88	1.39
os300_za_2022_n300	10	8.8	9.08	0.81	8.48	9.31	1.19

Table 24: Results on MS-LOP for Δ_{SP} metric and $m = 15$ on os300 instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
os300_at_2022_n300	15	14.09	14.09	1.36	11.82	14.01	2.05
os300_au_2022_n300	15	9.83	13.65	2.12	14.99	13.32	2.12
os300_be_2022_n300	15	14.36	13.92	1.95	11.54	14.7	1.80
os300_bg_2022_n300	15	14.69	14.95	0.77	14.98	14.76	1.73
os300_br_2022_n300	15	14.28	14.39	0.95	15.0	13.49	2.09
os300_ca_2022_n300	15	14.99	14.22	1.99	11.57	13.61	2.08
os300_ch_2022_n300	15	12.39	14.37	1.48	14.97	14.81	1.82
os300_cn_2022_n300	15	13.78	13.78	2.42	12.84	13.94	2.08
os300_cy_2022_n300	15	14.99	14.93	0.63	9.48	14.66	1.98
os300_cz_2022_n300	15	14.95	14.37	1.66	14.93	14.78	2.11
os300_de_2022_n300	15	13.14	14.41	1.48	15.0	14.92	1.80
os300_dk_2022_n300	15	13.49	13.77	2.04	14.99	13.92	1.81
os300_ee_2022_n300	15	11.92	14.24	1.50	14.92	14.96	0.08
os300_es_2022_n300	15	13.49	14.69	1.17	12.9	13.92	1.82
os300_fi_2022_n300	15	10.98	14.38	2.03	14.99	14.75	2.14
os300_fr_2022_n300	15	14.09	14.68	1.05	13.2	14.34	1.71
os300_gb_2022_n300	15	14.07	14.39	1.08	14.99	14.87	2.09
os300_gr_2022_n300	15	14.98	13.75	1.77	14.95	14.05	2.05
os300_hr_2022_n300	15	14.98	14.23	1.22	15.0	14.3	2.09
os300_hu_2022_n300	15	14.84	14.48	1.12	14.38	14.81	1.72
os300_id_2022_n300	15	14.69	14.52	1.20	11.58	14.51	2.01
os300_ie_2022_n300	15	13.19	13.04	1.82	10.5	12.42	2.42
os300_in_2022_n300	15	12.86	13.85	1.28	12.09	13.19	2.30
os300_it_2022_n300	15	15.0	14.81	1.19	14.98	14.07	2.10
os300_jp_2022_n300	15	14.94	13.8	1.79	15.0	14.49	1.80
os300_kr_2022_n300	15	14.01	14.08	2.18	12.86	13.2	2.09
os300_lt_2022_n300	15	14.05	14.19	1.51	15.0	14.91	1.80
os300_lu_2022_n300	15	14.39	13.18	1.49	15.0	13.16	2.87
os300_lv_2022_n300	15	14.97	14.96	0.97	15.0	14.92	1.83
os300_mt_2022_n300	15	14.99	14.07	1.52	11.52	13.19	2.99
os300_mx_2022_n300	15	14.88	13.79	1.88	15.0	13.8	2.05
os300_nl_2022_n300	15	13.49	13.74	1.72	14.93	14.88	1.85
os300_no_2022_n300	15	14.73	14.09	1.52	14.99	13.14	2.82
os300_pl_2022_n300	15	14.93	14.67	1.12	14.01	14.36	1.81
os300_pt_2022_n300	15	12.05	14.37	1.88	14.98	14.82	1.82
os300_ro_2022_n300	15	14.93	13.8	1.51	13.46	14.63	2.02
os300_ru_2022_n300	15	14.93	13.79	1.49	13.19	14.62	1.79
os300_se_2022_n300	15	15.0	14.53	1.20	14.92	14.96	1.12
os300_si_2022_n300	15	14.93	14.83	1.12	14.99	14.94	0.70
os300_sk_2022_n300	15	12.34	13.75	1.46	14.99	14.61	1.79
os300_tr_2022_n300	15	11.78	14.5	1.11	14.04	13.94	2.04
os300_tw_2022_n300	15	13.78	14.08	1.45	12.09	14.66	1.70
os300_us_2022_n300	15	14.38	14.09	1.11	12.88	13.94	2.84
os300_wa_2022_n300	15	14.98	14.08	1.63	12.82	14.24	1.91
os300_we_2022_n300	15	14.98	14.89	1.20	14.94	14.96	0.06
os300_wf_2022_n300	15	14.99	14.72	1.00	14.07	14.76	1.80
os300_wl_2022_n300	15	13.77	14.06	1.81	14.96	13.46	2.00
os300_wm_2022_n300	15	13.74	14.05	1.61	11.57	13.46	2.10
os300_za_2022_n300	15	13.49	13.78	1.20	12.58	13.92	1.98

Table 25: Results on MS-LOP for Δ_{SP} metric and $m = 5$ on PXP instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
pxp_at_2022_n155	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_au_2022_n155	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_be_2022_n155	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_bg_2022_n135	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_br_2022_n155	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_ca_2022_n158	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_ch_2022_n141	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_cn_2022_n169	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_cy_2022_n116	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_cz_2022_n162	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_de_2022_n165	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_dk_2022_n141	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_ee_2022_n126	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_es_2022_n162	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_fi_2022_n153	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_fr_2022_n164	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_gb_2022_n159	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_gr_2022_n130	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_hr_2022_n140	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_hu_2022_n160	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_id_2022_n132	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_ie_2022_n136	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_in_2022_n145	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_it_2022_n165	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_jp_2022_n159	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_kr_2022_n149	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_lt_2022_n133	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_lu_2022_n120	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_lv_2022_n132	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_mt_2022_n89	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_mx_2022_n139	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_nl_2022_n158	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_no_2022_n145	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_pl_2022_n166	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_pt_2022_n149	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_ro_2022_n151	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_ru_2022_n136	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_se_2022_n156	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_si_2022_n129	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_sk_2022_n151	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_tr_2022_n145	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_tw_2022_n138	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_us_2022_n170	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_wa_2022_n176	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_we_2022_n144	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_wf_2022_n141	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_wl_2022_n164	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_wm_2022_n161	5	5.0	5.0	0.00	5.0	5.0	0.00
pxp_za_2022_n134	5	5.0	5.0	0.00	5.0	5.0	0.00

Table 26: Results on MS-LOP for Δ_{SP} metric and $m = 10$ on PXP instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
pxp_at_2022_n155	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_au_2022_n155	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_be_2022_n155	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_bg_2022_n135	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_br_2022_n155	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_ca_2022_n158	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_ch_2022_n141	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_cn_2022_n169	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_cy_2022_n116	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_cz_2022_n162	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_de_2022_n165	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_dk_2022_n141	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_ee_2022_n126	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_es_2022_n162	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_fi_2022_n153	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_fr_2022_n164	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_gb_2022_n159	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_gr_2022_n130	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_hr_2022_n140	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_hu_2022_n160	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_id_2022_n132	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_ie_2022_n136	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_in_2022_n145	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_it_2022_n165	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_jp_2022_n159	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_kr_2022_n149	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_lt_2022_n133	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_lu_2022_n120	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_lv_2022_n132	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_mt_2022_n89	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_mx_2022_n139	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_nl_2022_n158	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_no_2022_n145	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_pl_2022_n166	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_pt_2022_n149	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_ro_2022_n151	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_ru_2022_n136	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_se_2022_n156	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_si_2022_n129	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_sk_2022_n151	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_tr_2022_n145	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_tw_2022_n138	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_us_2022_n170	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_wa_2022_n176	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_we_2022_n144	10	10.0	10.0	0.01	10.0	10.0	0.00
pxp_wf_2022_n141	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_wl_2022_n164	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_wm_2022_n161	10	10.0	10.0	0.00	10.0	10.0	0.00
pxp_za_2022_n134	10	10.0	10.0	0.00	10.0	10.0	0.00

Table 27: Results on MS-LOP for Δ_{SP} metric and $m = 15$ on PXP instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
pxp_at_2022_n155	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_au_2022_n155	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_be_2022_n155	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_bg_2022_n135	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_br_2022_n155	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_ca_2022_n158	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_ch_2022_n141	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_cn_2022_n169	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_cy_2022_n116	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_cz_2022_n162	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_de_2022_n165	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_dk_2022_n141	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_ee_2022_n126	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_es_2022_n162	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_fi_2022_n153	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_fr_2022_n164	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_gb_2022_n159	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_gr_2022_n130	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_hr_2022_n140	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_hu_2022_n160	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_id_2022_n132	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_ie_2022_n136	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_in_2022_n145	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_it_2022_n165	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_jp_2022_n159	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_kr_2022_n149	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_lt_2022_n133	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_lu_2022_n120	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_lv_2022_n132	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_mt_2022_n89	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_mx_2022_n139	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_nl_2022_n158	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_no_2022_n145	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_pl_2022_n166	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_pt_2022_n149	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_ro_2022_n151	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_ru_2022_n136	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_se_2022_n156	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_si_2022_n129	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_sk_2022_n151	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_tr_2022_n145	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_tw_2022_n138	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_us_2022_n170	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_wa_2022_n176	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_we_2022_n144	15	14.98	14.96 ▽	0.12	15.0	15.0 ▲	0.00
pxp_wf_2022_n141	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_wl_2022_n164	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_wm_2022_n161	15	15.0	15.0	0.00	15.0	15.0	0.00
pxp_za_2022_n134	15	15.0	15.0	0.00	15.0	15.0	0.00

Table 28: Results on MS-LOP for Δ_{SP} metric and $m = 5$ on RXR instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
rxr_1995_n49	5	4.97	4.97	0.00	4.97	4.97	0.00
rxr_1996_n49	5	5.0	5.0	0.00	5.0	5.0	0.00
rxr_1997_n49	5	4.4	4.4	0.30	4.4	4.4	0.00
rxr_1998_n49	5	5.0	5.0	0.00	5.0	5.0	0.00
rxr_1999_n49	5	4.96	4.96	0.00	4.96	4.96	0.00
rxr_2000_n49	5	4.69	4.69	0.00	4.69	4.69	0.00
rxr_2001_n49	5	4.4	4.4	0.00	4.4	4.4	0.00
rxr_2002_n49	5	5.0	5.0	0.00	5.0	5.0	0.00
rxr_2003_n49	5	4.98	4.98	0.00	4.98	4.98	0.00
rxr_2004_n49	5	5.0	5.0	0.00	5.0	5.0	0.00
rxr_2005_n49	5	5.0	4.99	0.01	4.99	4.99	0.00
rxr_2006_n49	5	4.7	4.7	0.30	4.7	4.7	0.00
rxr_2007_n49	5	5.0	5.0	0.00	5.0	5.0	0.00
rxr_2008_n49	5	5.0	5.0	0.00	5.0	5.0	0.00
rxr_2009_n49	5	4.4	4.4	0.00	4.4	4.4	0.00
rxr_2010_n49	5	5.0	5.0	0.00	5.0	5.0	0.00
rxr_2011_n49	5	5.0	5.0	1.00	5.0	5.0	0.00
rxr_2012_n49	5	5.0	5.0	0.00	5.0	5.0	0.00
rxr_2013_n49	5	5.0	5.0	0.00	5.0	5.0	0.00
rxr_2014_n49	5	5.0	5.0	0.00	5.0	5.0	0.00
rxr_2015_n49	5	4.99	4.99	0.00	4.99	4.99	0.00
rxr_2016_n49	5	4.7	4.7 $\blacktriangle$	0.00	4.4	4.4 ∇	0.00
rxr_2017_n49	5	5.0	5.0 $\blacktriangle$	0.00	4.0	4.0 ∇	1.75
rxr_2018_n49	5	5.0	5.0	0.00	5.0	5.0	0.00
rxr_2019_n49	5	3.0	3.0 $\blacktriangle$	0.75	2.0	2.0 ∇	0.00
rxr_2020_n49	5	5.0	5.0	0.00	5.0	5.0	0.00
rxr_2021_n49	5	5.0	4.5 ∇	1.00	5.0	5.0 $\blacktriangle$	0.00
rxr_2022_n49	5	4.99	4.99	0.00	4.99	4.99	0.00

Table 29: Results on MS-LOP for Δ_{SP} metric and $m = 10$ on RXR instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
rxr_1995_n49	10	9.97	9.96	0.00	9.96	9.96	0.00
rxr_1996_n49	10	5.0	5.0 ∇	1.00	7.0	7.0 $\blacktriangle$	1.00
rxr_1997_n49	10	9.4	9.1 $\blacktriangle$	0.90	8.5	8.5 ∇	0.30
rxr_1998_n49	10	8.0	8.0 $\blacktriangle$	1.00	5.0	6.0 ∇	1.00
rxr_1999_n49	10	9.92	9.92 $\blacktriangle$	0.01	9.9	9.9 ∇	0.00
rxr_2000_n49	10	9.07	9.07	0.00	9.07	9.07	0.30
rxr_2001_n49	10	9.09	9.09	0.00	9.09	9.09	0.00
rxr_2002_n49	10	10.0	10.0	0.00	10.0	10.0	0.00
rxr_2003_n49	10	8.97	6.98 $\blacktriangle$	0.99	5.98	5.98 ∇	1.00
rxr_2004_n49	10	10.0	10.0	0.00	10.0	10.0	0.00
rxr_2005_n49	10	9.99	9.99	0.00	9.99	9.98	0.01
rxr_2006_n49	10	5.7	6.7 ∇	1.30	8.4	7.75 $\blacktriangle$	1.00
rxr_2007_n49	10	7.0	6.0 $\blacktriangle$	1.00	6.0	5.0 ∇	1.00
rxr_2008_n49	10	10.0	10.0	0.00	10.0	10.0	0.00
rxr_2009_n49	10	9.09	8.79 $\blacktriangle$	0.60	8.49	8.49 ∇	0.00
rxr_2010_n49	10	9.0	9.0 $\blacktriangle$	1.00	7.0	8.0 ∇	1.00
rxr_2011_n49	10	5.0	5.0 ∇	1.00	6.0	6.0 $\blacktriangle$	0.00
rxr_2012_n49	10	6.0	6.0	1.00	5.0	6.0	1.00
rxr_2013_n49	10	10.0	10.0	0.00	9.0	10.0	1.00
rxr_2014_n49	10	6.0	7.0 ∇	1.00	10.0	10.0 $\blacktriangle$	0.00
rxr_2015_n49	10	9.95	9.96 $\blacktriangle$	0.74	9.97	7.97 ∇	2.00
rxr_2016_n49	10	7.7	7.7 ∇	1.62	8.8	8.8 $\blacktriangle$	0.00
rxr_2017_n49	10	9.0	9.0 $\blacktriangle$	1.00	4.0	4.0 ∇	1.75
rxr_2018_n49	10	10.0	10.0 $\blacktriangle$	0.23	9.7	9.7 ∇	0.53
rxr_2019_n49	10	3.0	3.0 $\blacktriangle$	0.75	2.0	2.0 ∇	0.00
rxr_2020_n49	10	10.0	10.0	0.00	10.0	10.0	0.00
rxr_2021_n49	10	5.0	4.5 ∇	1.00	6.0	5.0 $\blacktriangle$	1.00
rxr_2022_n49	10	9.98	9.98	0.00	9.98	9.98	0.00

Table 30: Results on MS-LOP for Δ_{SP} metric and $m = 15$ on RXR instances

Instance	m	MS-CD-RVNS			MS-MA-EDM		
		Best	Median	IQR	Best	Median	IQR
rxr_1995_n49	15	14.95	14.95 $\blacktriangle$	0.01	14.94	14.94 ∇	0.00
rxr_1996_n49	15	5.0	5.0 ∇	1.00	7.0	7.0 $\blacktriangle$	1.00
rxr_1997_n49	15	14.4	13.5 $\blacktriangle$	0.50	12.9	12.9 ∇	0.00
rxr_1998_n49	15	8.0	8.0 $\blacktriangle$	1.00	5.0	6.0 ∇	1.00
rxr_1999_n49	15	14.9	14.91 $\blacktriangle$	0.02	14.89	14.89 ∇	0.01
rxr_2000_n49	15	14.04	13.74 $\blacktriangle$	0.30	13.74	13.45 ∇	0.29
rxr_2001_n49	15	14.09	14.09 $\blacktriangle$	0.00	9.09	12.09 ∇	2.00
rxr_2002_n49	15	15.0	15.0	0.00	15.0	15.0	0.00
rxr_2003_n49	15	8.97	6.98 $\blacktriangle$	0.99	5.98	5.98 ∇	1.00
rxr_2004_n49	15	15.0	15.0 $\blacktriangle$	0.00	15.0	14.5 ∇	2.00
rxr_2005_n49	15	14.99	14.99 $\blacktriangle$	0.01	11.99	12.48 ∇	2.00
rxr_2006_n49	15	5.7	6.7 ∇	1.30	8.4	7.75 $\blacktriangle$	1.00
rxr_2007_n49	15	7.0	6.0 $\blacktriangle$	1.00	6.0	5.0 ∇	1.00
rxr_2008_n49	15	15.0	15.0	0.00	15.0	15.0	0.00
rxr_2009_n49	15	13.49	13.19 $\blacktriangle$	1.13	12.89	12.89 ∇	0.00
rxr_2010_n49	15	9.0	9.0 $\blacktriangle$	1.00	7.0	8.0 ∇	1.00
rxr_2011_n49	15	5.0	5.0 ∇	1.00	6.0	6.0 $\blacktriangle$	0.00
rxr_2012_n49	15	6.0	6.0	1.00	5.0	6.0	1.00
rxr_2013_n49	15	13.0	13.0 $\blacktriangle$	1.00	9.0	10.0 ∇	1.00
rxr_2014_n49	15	6.0	7.0 ∇	1.00	11.0	10.0 $\blacktriangle$	1.00
rxr_2015_n49	15	11.95	9.97 $\blacktriangle$	2.73	9.97	7.97 ∇	2.00
rxr_2016_n49	15	7.7	7.7 ∇	1.62	13.2	12.9 $\blacktriangle$	1.70
rxr_2017_n49	15	9.0	9.0 $\blacktriangle$	1.00	4.0	4.0 ∇	1.75
rxr_2018_n49	15	15.0	14.7 $\blacktriangle$	0.30	12.7	11.35 ∇	2.93
rxr_2019_n49	15	3.0	3.0 $\blacktriangle$	0.75	2.0	2.0 ∇	0.00
rxr_2020_n49	15	14.4	14.4 ∇	0.48	15.0	15.0 $\blacktriangle$	0.00
rxr_2021_n49	15	5.0	4.5 ∇	1.00	6.0	5.0 $\blacktriangle$	1.00
rxr_2022_n49	15	14.98	14.98 $\blacktriangle$	0.00	9.98	11.98 ∇	1.99



Figure 1: Φ vs. Δ_{NN} for all the os300 instances with $m = 5$.

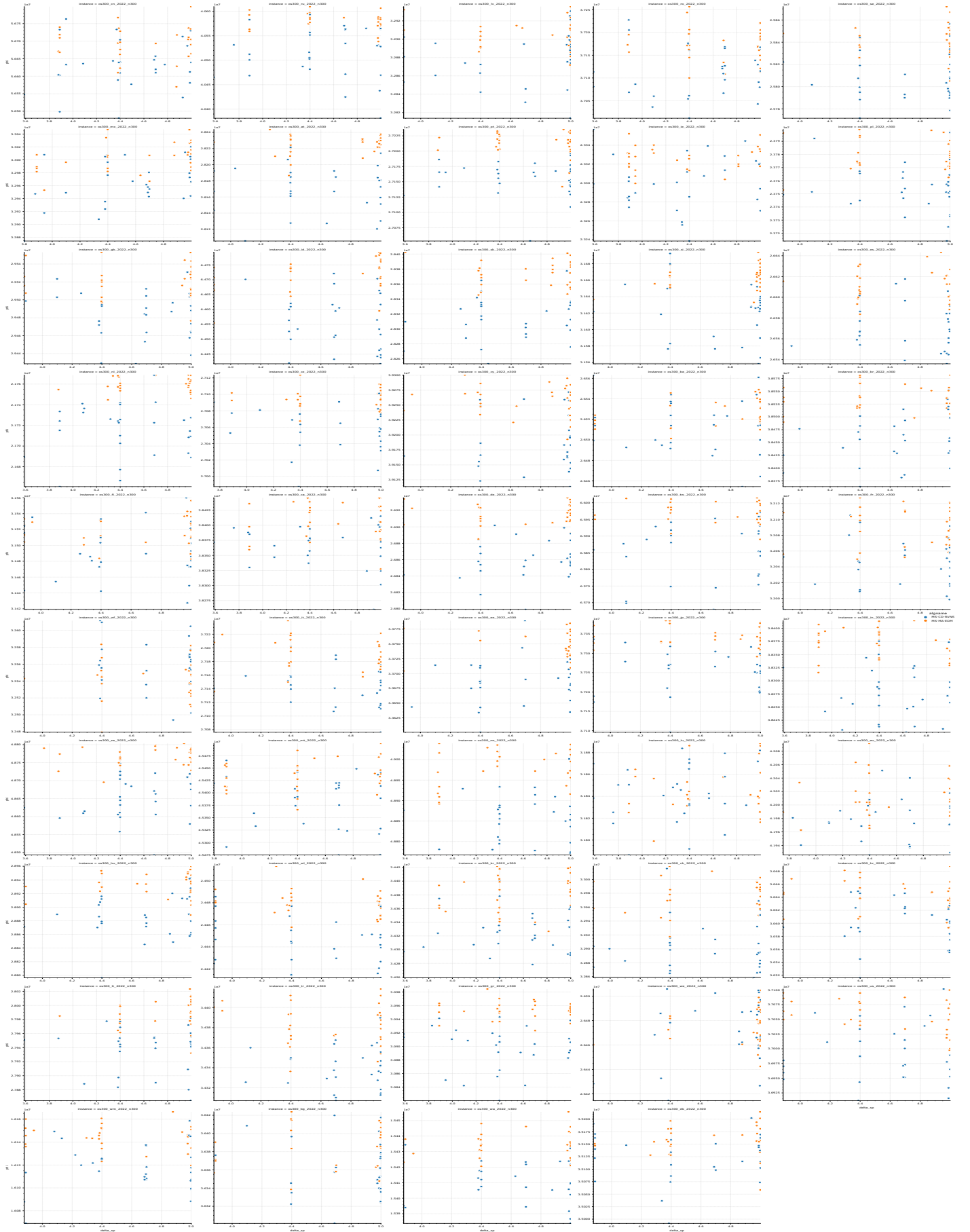


Figure 2: Φ vs. Δ_{SP} for all the os300 instances with $m = 5$.

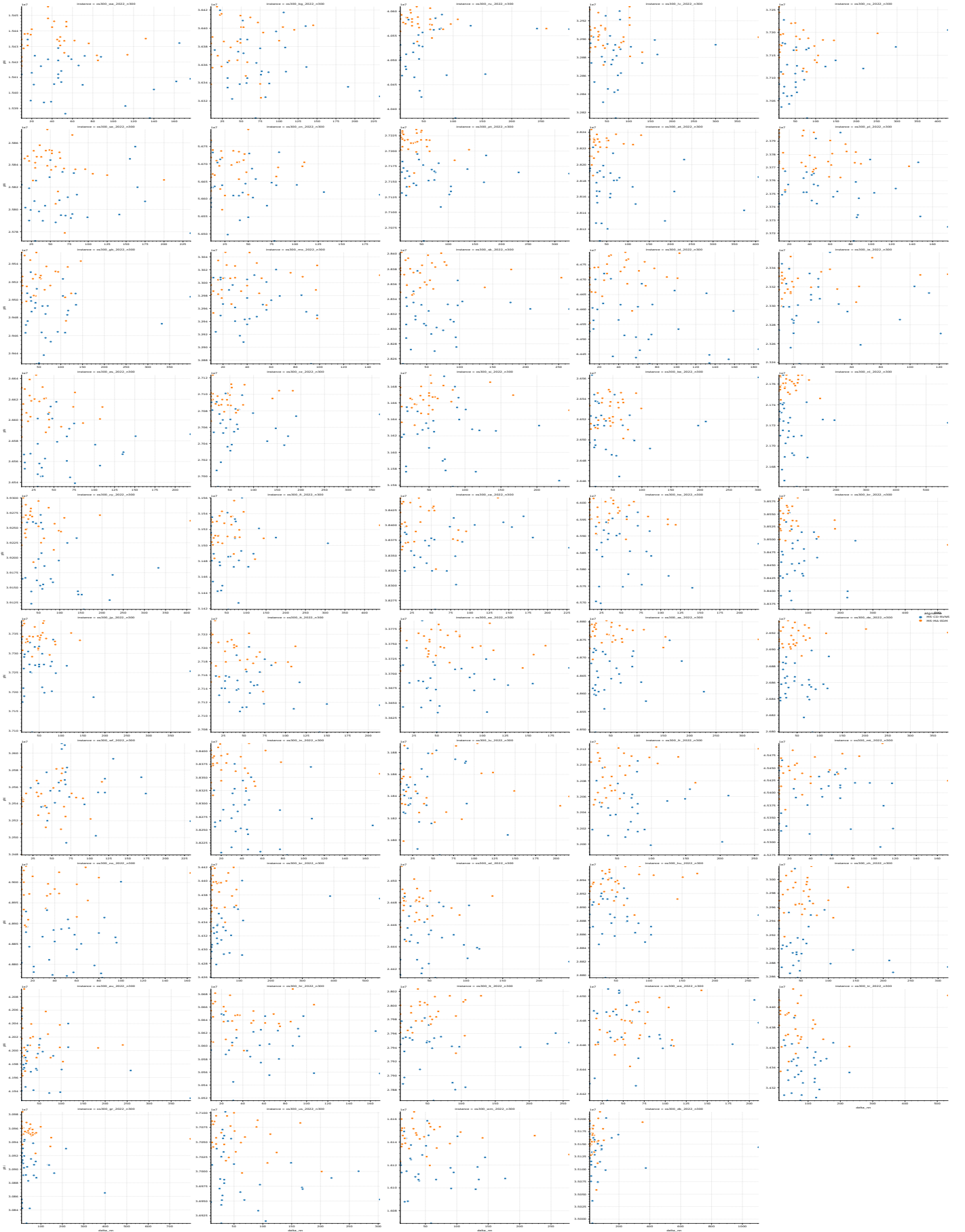


Figure 3: Φ vs. Δ_N for all the os300 instances with $m = 10$.

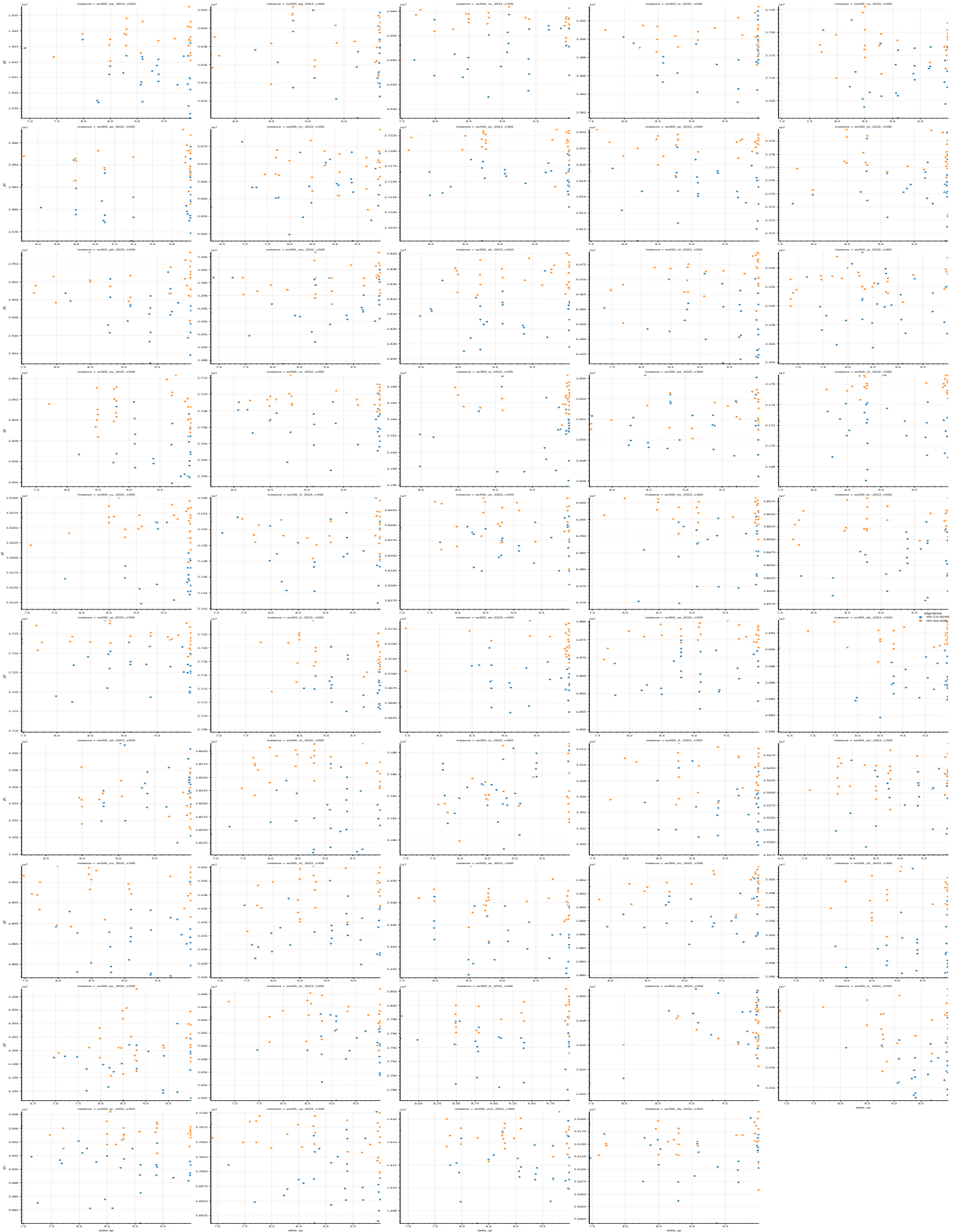


Figure 4: Φ vs. Δ_{SP} for all the os300 instances with $m = 10$.

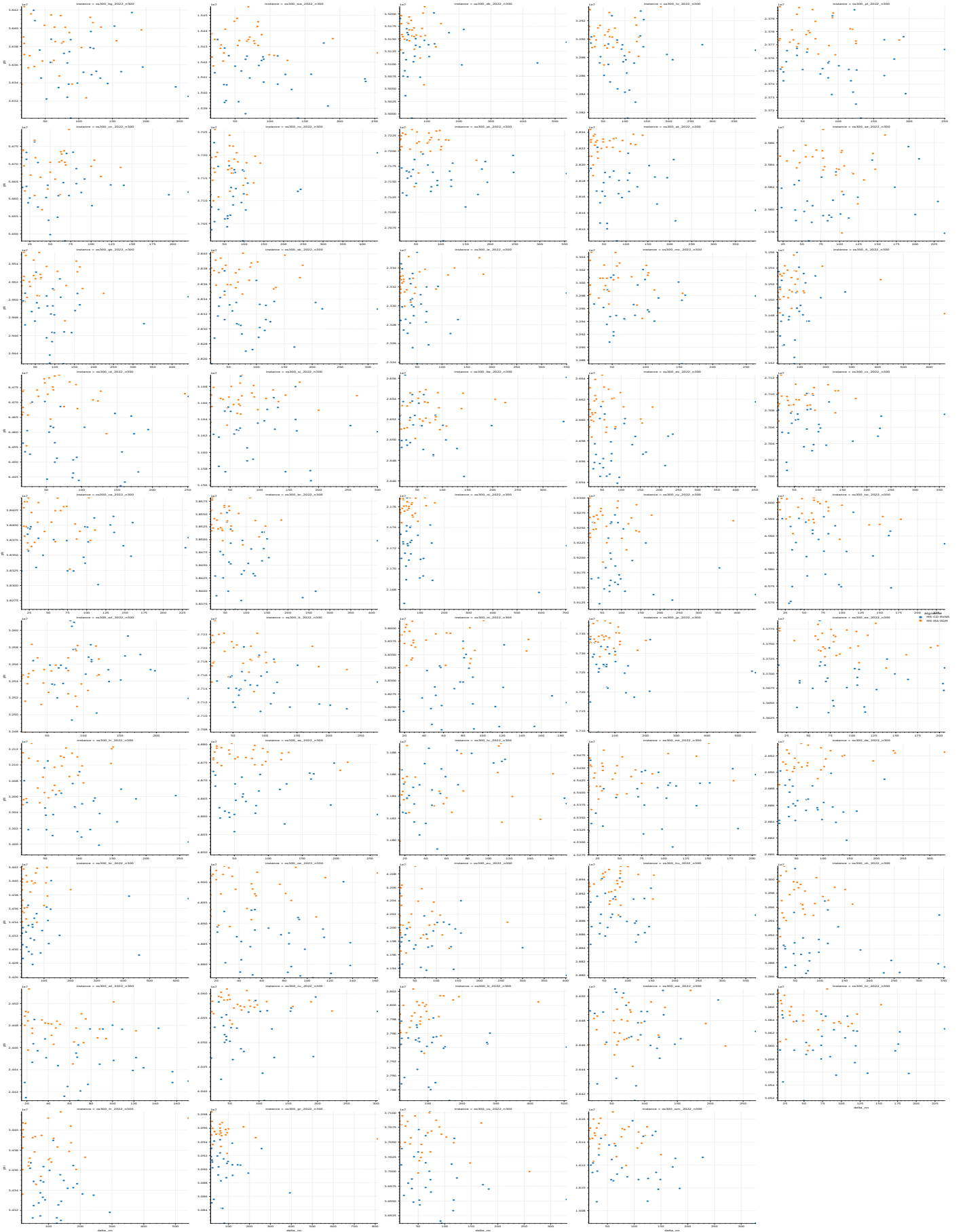


Figure 5: Φ vs. Δ_N for all the os300 instances with $m = 15$.

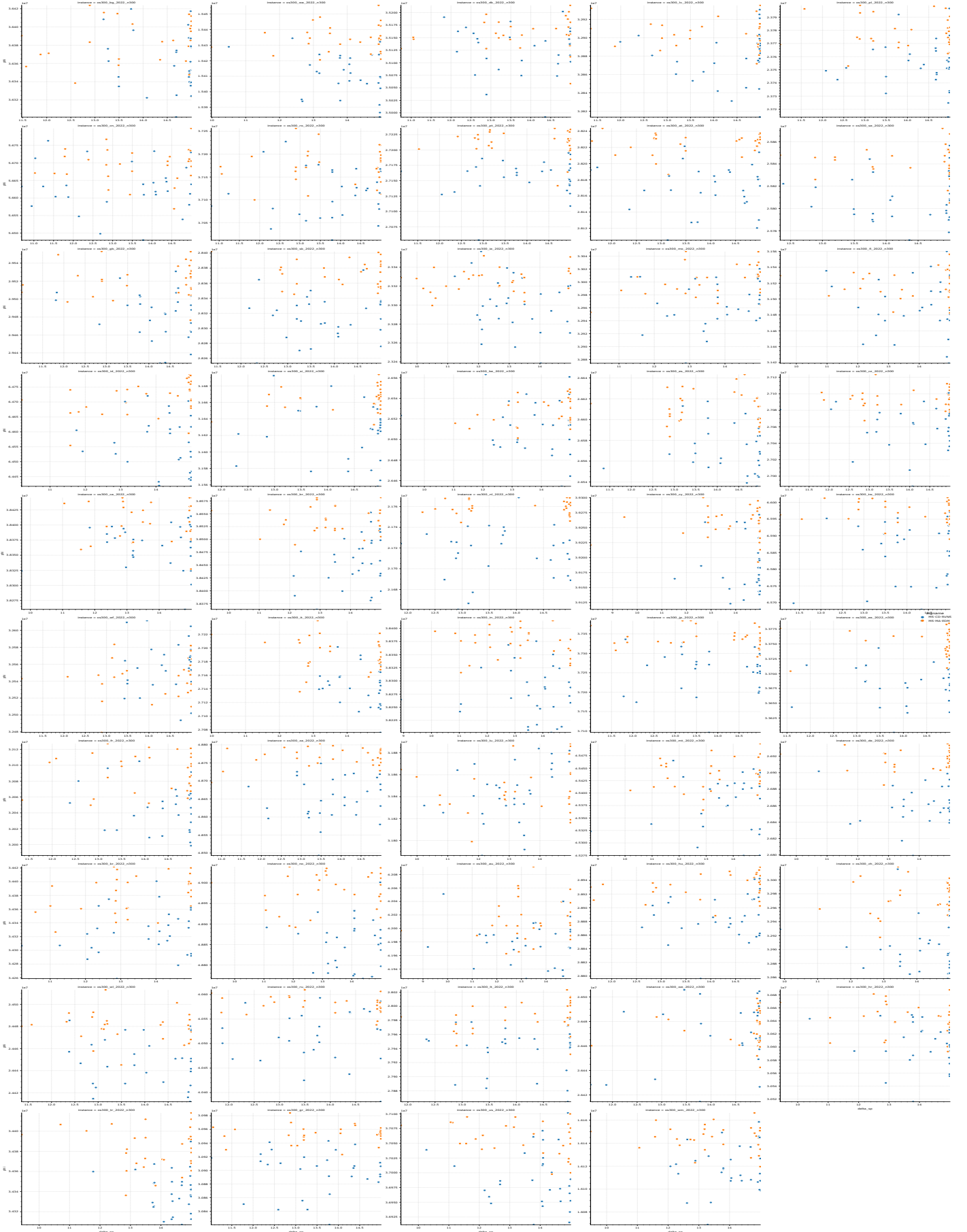


Figure 6: Φ vs. Δ_{SP} for all the os300 instances with $m = 15$.



Figure 7: Φ vs. Δ_{NN} for all the pxp instances with $m = 5$.

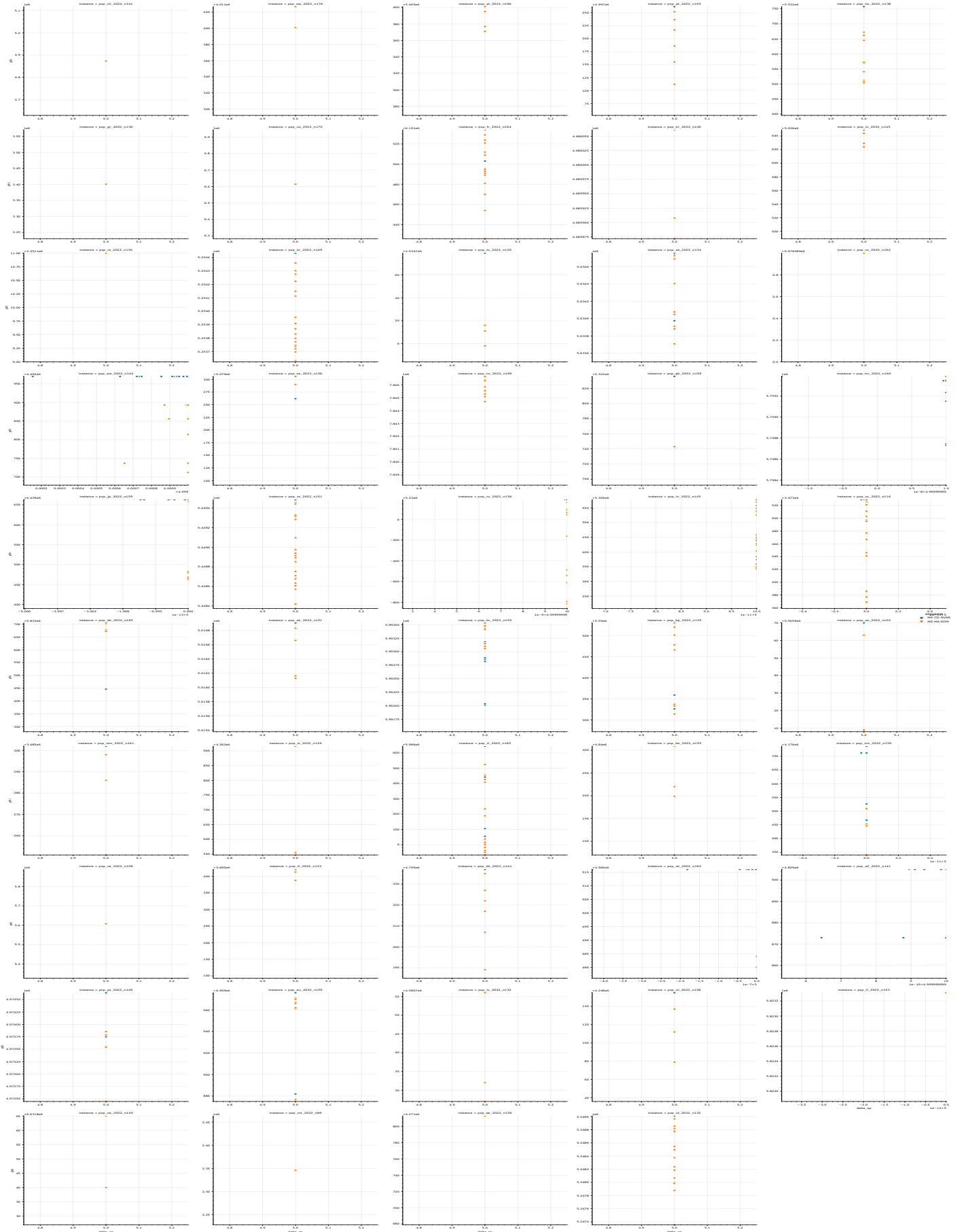


Figure 8: Φ vs. Δ_{SP} for all the pxp instances with $m = 5$.



Figure 9: Φ vs. Δ_{NN} for all the pxp instances with $m = 10$.



Figure 11: Φ vs. Δ_{NN} for all the pxp instances with $m = 15$.

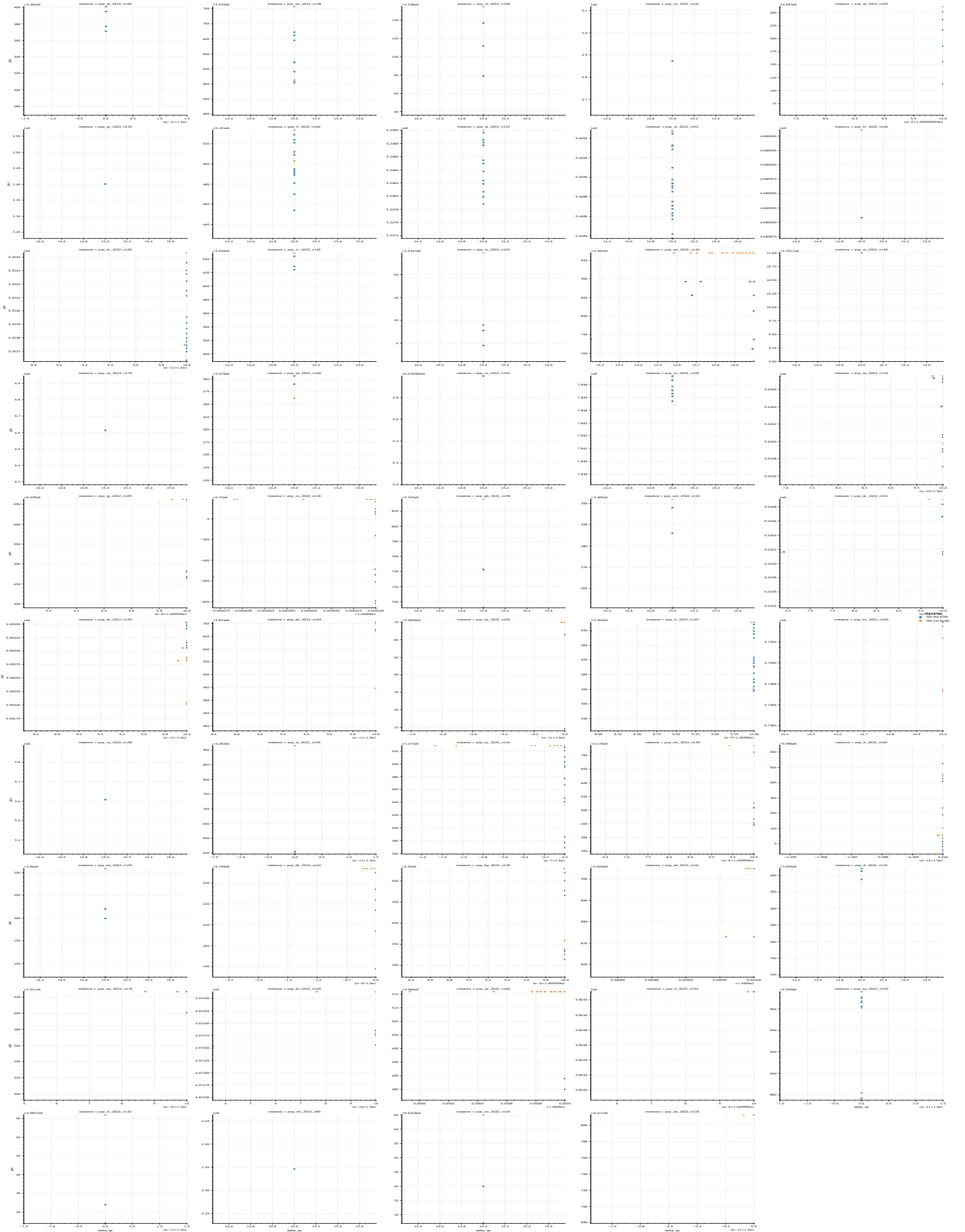


Figure 12: Φ vs. Δ_{SP} for all the ppx instances with $m = 15$.

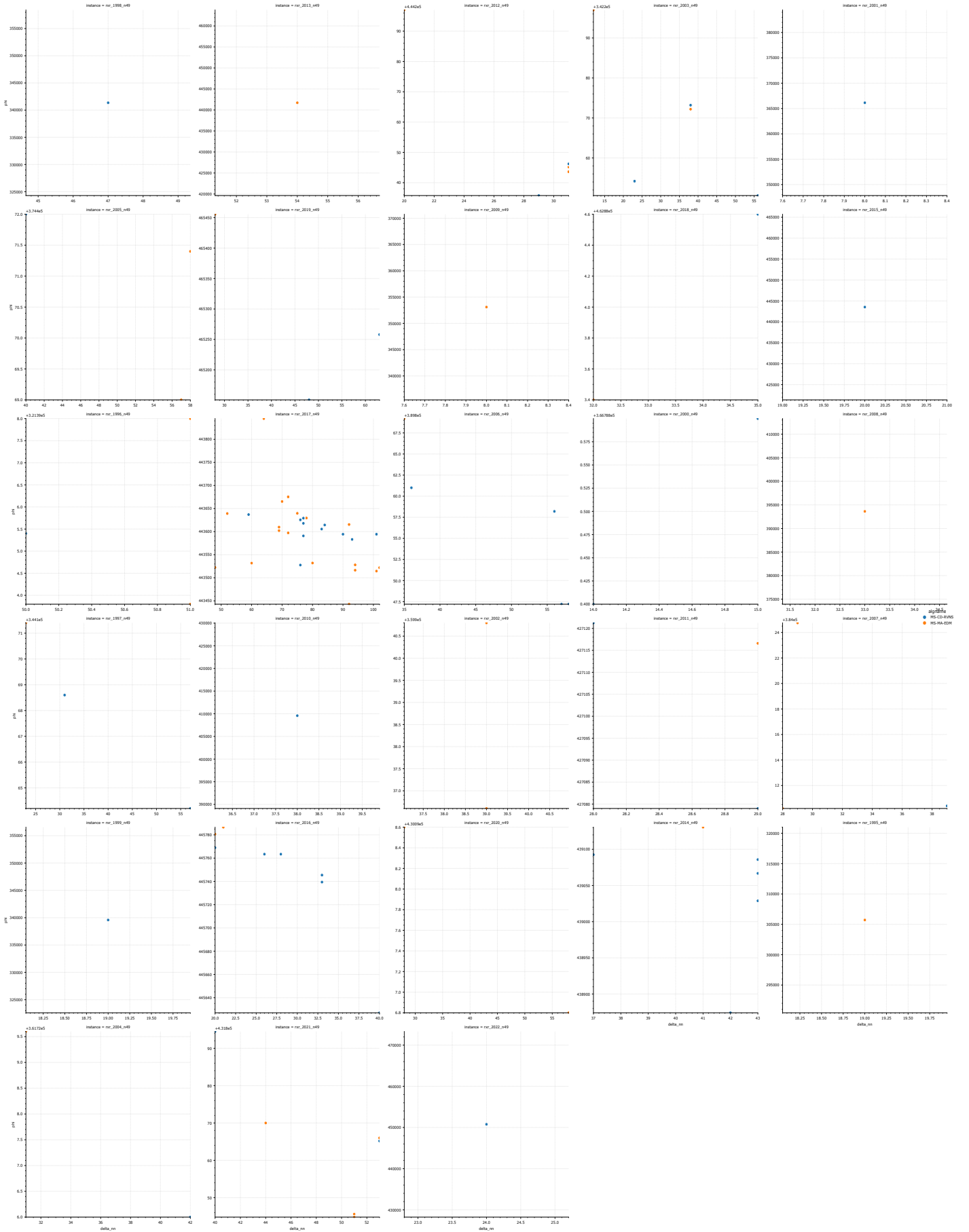


Figure 13: Φ vs. Δ_{NN} for all the rxr instances with $m = 5$.

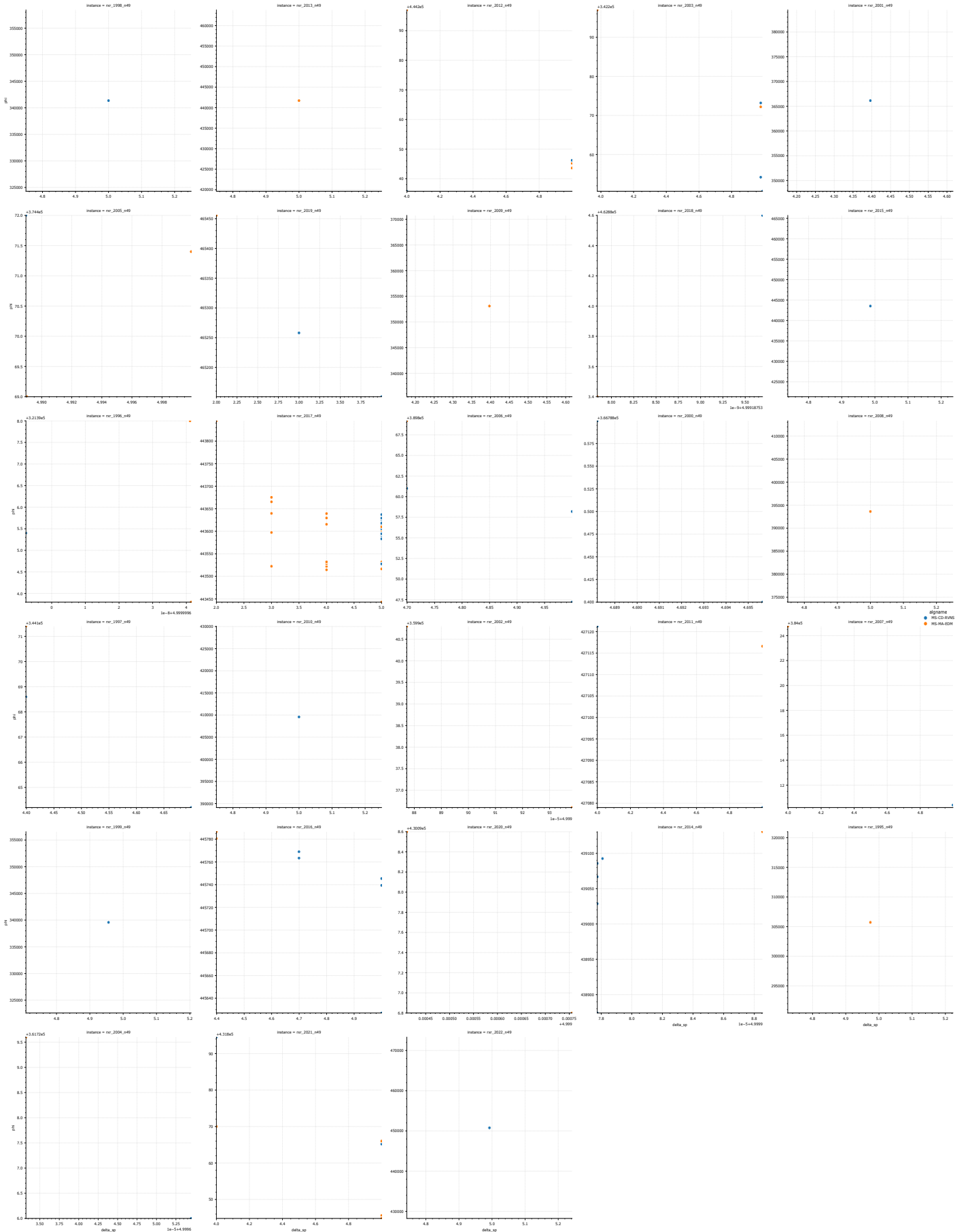


Figure 14: Φ vs. Δ_{SP} for all the rxr instances with $m = 5$.

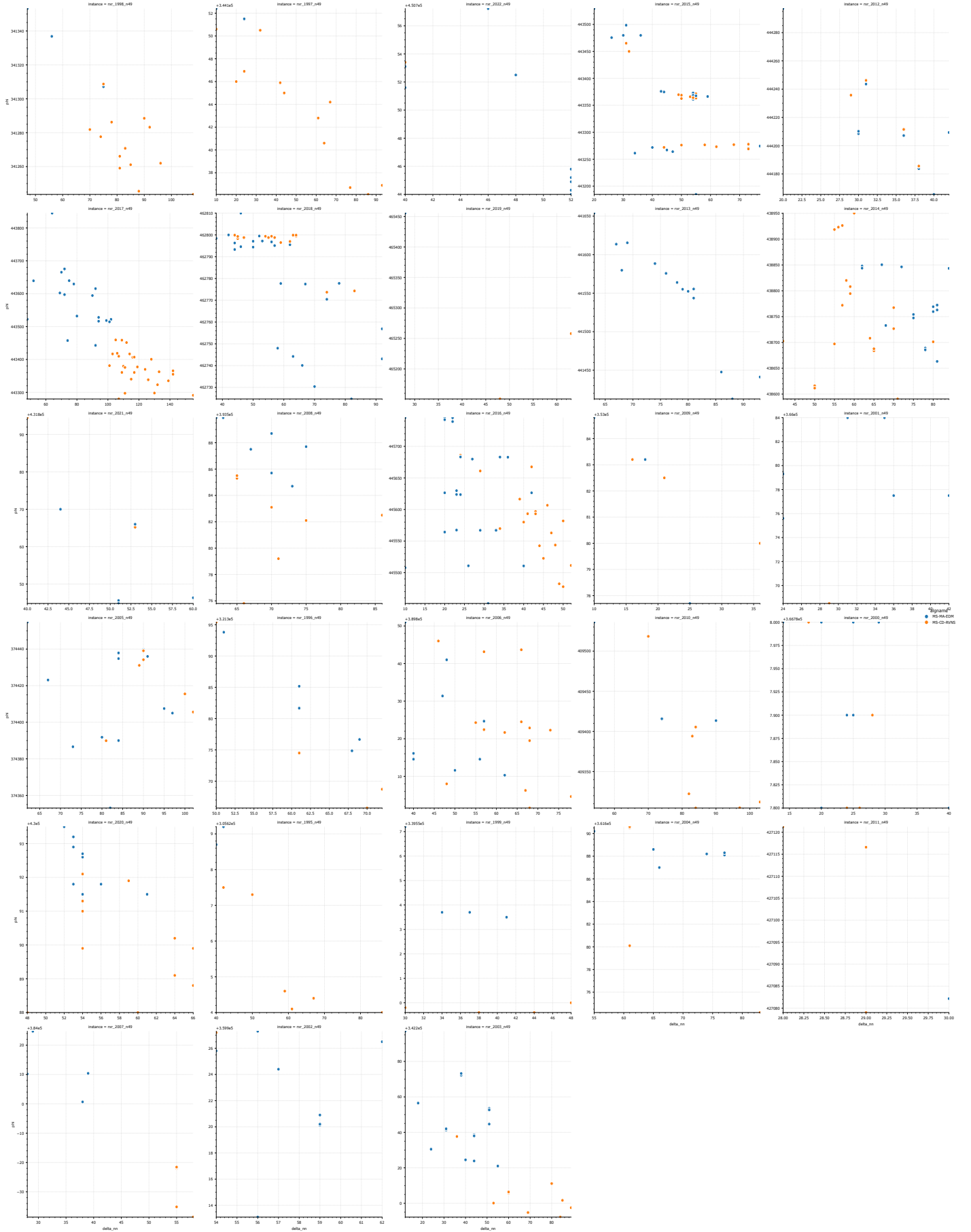


Figure 15: Φ vs. Δ_{NN} for all the rxr instances with $m = 10$.

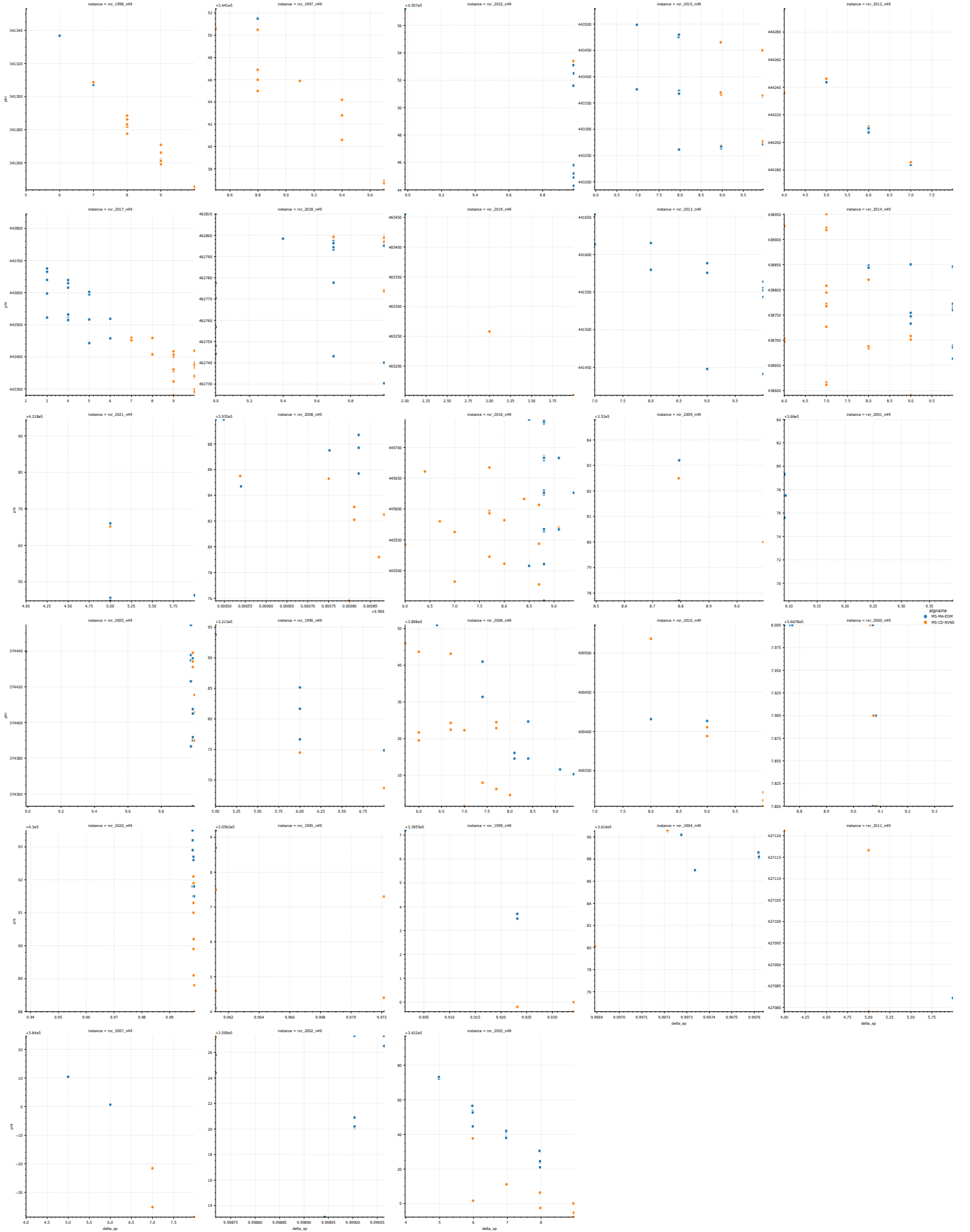


Figure 16: Φ vs. Δ_{SP} for all the rxr instances with $m = 10$.



Figure 17: Φ vs. Δ_{NN} for all the rxr instances with $m = 15$.

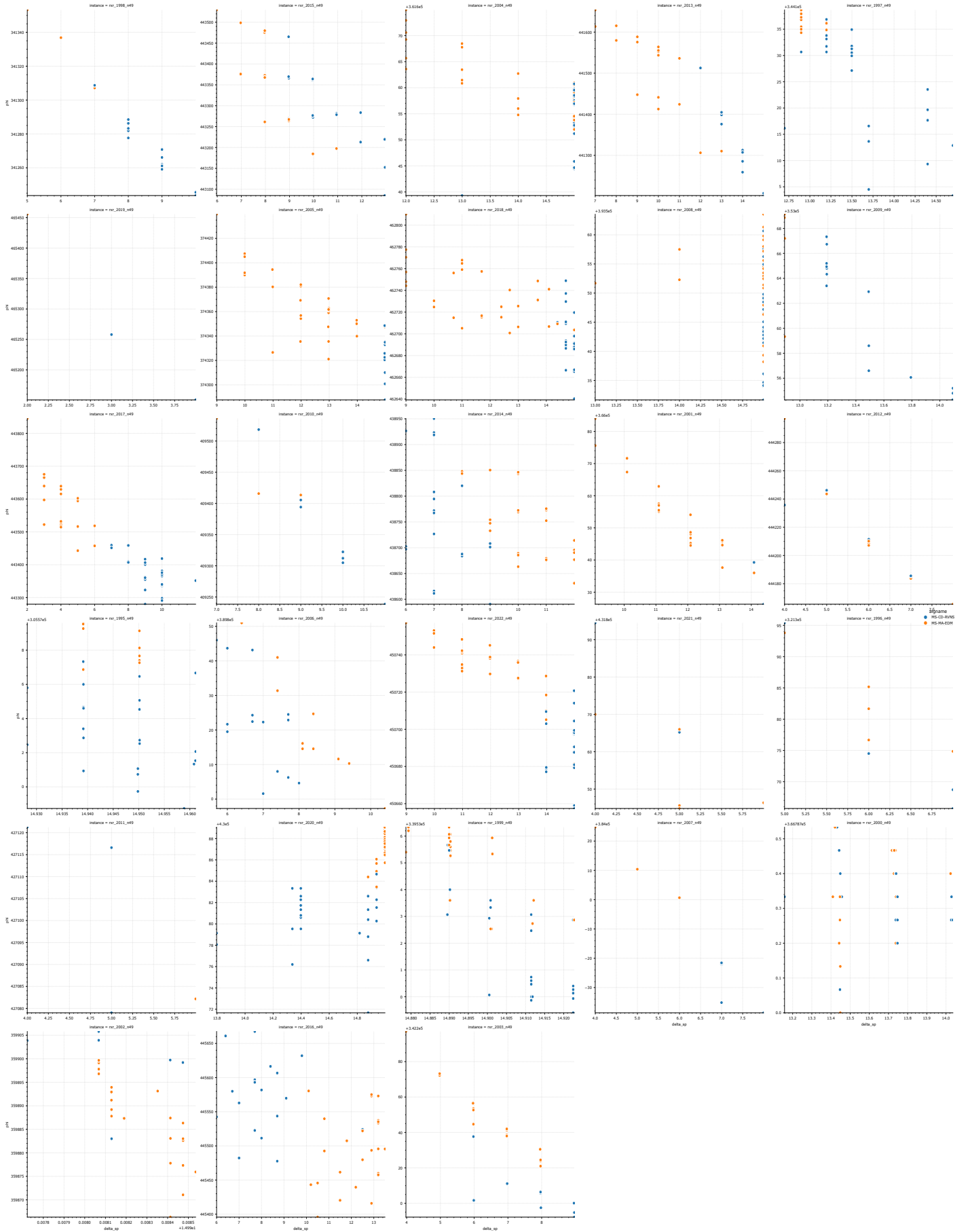


Figure 18: Φ vs. Δ_{SP} for all the rxr instances with $m = 15$.